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Healthcare related needs and desires of people with post-stroke aphasia: A "double meta-synthesis" using two different aproaches – study protocol

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Healthcare related needs and desires of people with post-stroke aphasia: A "double meta-synthesis" using two different aproaches – study protocol

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Keywords: qualitative research, qualitative synthesis, meta-ethnography, thematic synthesis, aphasia

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Abstract

Introduction: Stroke is a frequent disease in the elderly population of Western European Countries with aphasia as a very common consequence. Aphasia is known to impede targeting treatment to individual needs of patients – and therefore may reduce treatment success. One of our central aims of the "double meta-synthesis" is to explore the distinct health care related needs and wishes of older people with poststroke aphasia. Since various methods of meta-synthesizing qualitative research exist, the aim of the study in which the meta-synthesises are embedded is to clarify, whether and how results differ according to the approach used.

Methods and analysis: We will conduct a "double meta-synthesis" by using meta-ethnography (ME) and thematic synthesis (ThS) independently to summarize the results of primary qualitative studies. We chose two methods with different underlying epistemologies [idealistic (ME) vs. a realistic (ThS)] because this has a significant impact on methodological issues.

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At least four databases will be searched. Synthesis will follow the meta-ethnographic/thematic synthesis approach. Inclusion criteria comprise: patients aged 65 and older, studies published in German/English and all types of qualitative studies concerning needs and desires related to health care or the health care system. The protocol is registered on PROSPERO and is reported using PRISMA-P guidelines and three items from the ENTREQ checklist. Results will not only give insights into needs and desires of patients with post-stroke aphasia but also contribute to the methodological discussion on meta-synthesis methods for qualitative research.

Ethics and dissemination: Ethical approval is not required. Results will be published in a peer-reviewed journal. They will inform policy and practice how to optimize care for people suffering from stroke-related aphasia and also answer the question if the use of two synthesis methods leads to different results.

PROSPERO registration number: (registration pending)

Article summary

Strengths and limitations of this study

- We will conduct the meta-synthesis in an interdisciplinary researcher team with members highly experienced in qualitative research.
- All steps of the research process will be discussed within our research team and with the interdisciplinary working group “Qualitative Methods in Healthcare Research” of the German Network of Healthcare Research.
- Study screening and data extraction will be conducted independently by teams of two authors, and a fifth author will mediate any disagreements.
- The inclusion of only English and German language publications could introduce potential for cultural bias in the findings.
- Grey literature will not be included in the meta-synthesis.

Introduction

Stroke is a frequent disease in western European countries. Busch et al. found a lifetime-prevalence of 2.9% in the age-group between 40 and 79 that did not change significantly over the preceding 12 years [1]. Aphasia is known to be a very common consequence of stroke: It occurs in about 30% of people with stroke [2]. No primary epidemiological data of prevalence and incidence of poststroke aphasia in Germany are available. According to Huber et al. [3] in Germany about 25.000 people are newly affected by aphasia each year. Aphasic symptoms disappear frequently in the short-term, but remain in about 56% of patients more than ten months after the stroke [4]. This has considerable impact on postacute treatment of stroke, because the patient’s reduced ability to communicate severely affects the interaction between patient and health care provider.

Several quantitative investigations explored whether stroke survivor’s needs are addressed adequately in the health care system [5–11]. All of them revealed a substantial proportion of individuals with unmet needs from at least 51% [9] up to

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100 % [8, 12]. These studies clearly show that existing unmet needs are related to the severity of disease or disability as well as psychosocial aspects, living in deprived areas, low income or being member of an ethnic minority group. Chen et al. [13] discovered, that on average 73.8% of stroke survivors have at least one unmet need. The types of needs mentioned in their systematic review ranged from needs related to body functions, activities and participation, or environment, to service needs in information, transport, home help / personal care and therapy.

Methods for synthesizing qualitative research: which one to choose when?

Our aim is not only to generate results with respect to the needs of people with poststroke aphasia but also to contribute to the methodological discussion on qualitative meta-synthesis. Albeit theory-based recommendations for deciding which type of synthesis to choose for certain qualitative research questions exist [14–16], to date there is no empirical evidence showing which approach serves best in qualitative health care research. Therefore, we chose meta-ethnography (ME) and thematic synthesis (ThS) out of the numerous existing approaches for synthesizing qualitative research [14] to explore if and in which way the method of synthesizing impacts its results.

Thus, our research comprises two steps: The first step that is described here in detail encompasses the synthesis of studies focussing our content-related research question using two different methods. In a second step we will construct a framework that permits a structured comparison of the similarities and differences of the results generated with two different methods. This second step will be described in detail elsewhere.

Objective

The aim of our “double meta-synthesis” is to collect a corpus of qualitative studies that answer the question “What are the needs and desires related to health care of people suffering from aphasia after stroke?”. The studies will be synthesized using two different methods.

Methods

The protocol was registered with the International Prospective Register of Systematic Reviews (PROSPERO) on 27/02/2020 (registration pending). In the event of protocol modifications, changes will be documented in PROSPERO. We will provide the date of each modification and the rationale for the change.

The reporting of this protocol is in accordance with the *Preferred Reporting Items for Systematic Reviews and Meta-Analyses Statement for Protocols* (PRISMA-P [17]) and three additional items (item 17-19) from the *Enhancing Transparency in Reporting the synthesis of Qualitative research* (ENTREQ) checklist [18]). The reporting of the final review will also be according to PRISMA and ENTREQ recommendations. Ethical approval is not required, because no human subjects or personal data are directly involved in our review.

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Search for similar review projects

To verify that our meta-synthesis is embedded in current research we conducted a search for published and unpublished qualitative synthesis on our topic. We searched Medline via PubMed for published and PROSPERO for unpublished qualitative synthesis. We found three reviews related to our research question [19–21]. The review by Pindus et al. focusses on primary and community care and does include all stroke patients independent of aphasia status as well as informal caregivers [19]. The point of departure of the review from Hilari et al. was substantially lower health-related quality of life levels in stroke patients suffering from aphasia [20]. They aimed at identifying prognostic factors of health-related quality of life. Therefore, the analysis of health care needs is not the focus of their study. The third review by Manning et al. deals with a similar target group as well[21]. However, they also have no focus on health care needs or health care experience but are interested more generally in factors contributing to personal recovery and living successfully. In addition, none of the three reviews compared different methods of qualitative synthesis.

PICoS

Out of numerous tools that can be used for transferring research questions into answerable questions, e.g. PICO, PICo, SPIDER and more [22], we chose the PICo tool [23]. PICo in qualitative research means: **P**opulation, **I**nterest, **Co**ntext. Our objective is to identify studies that explore health care needs and desires (interest) of people with poststroke aphasia (population) during postacute treatment (context). We used the PICo-scheme complemented by requirements for the study design to create search terms and inclusion criteria (so-called PICoS-scheme).

Search

Databases and search terms

At least four electronic databases (Medline, PubMed, CINAHL, PSYCH-Info) will be searched from inception up to the day of the search. Search terms will be derived from the PICo(S) scheme and will include terms describing the population (e.g. aphasia, poststroke), interest (e.g. needs, desires, wishes), and study design (e.g. qualitative research, qualitative trial, ethnography). Filters will be used to reduce search results to certain languages. The search will be conducted by an expert for database search.

Additional search strategies

In addition to the database search, the review authors will screen the reference lists of included studies. No handsearching or forward-tracking strategy is planned.

Inclusion and exclusion criteria

We will use the criteria outlined in table 1 to select appropriate studies. As health care needs differ according to age [24] and most people affected by aphasia after stroke are of higher age [1], our study focuses on people aged 65 and older. Due to available resources we restricted publication language to English and German.

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Table 1: Inclusion and exclusion criteria for study selection

	Inclusion criteria	Exclusion criteria
Population	Adult People, aged 65 and older, with post-stroke aphasia	Study includes only members of particular (minority) groups
Interest of the study	Needs and desires related to (the) health care (system)	Study focusses special needs related to technical devices
		Studies examining certain therapeutic means
Context of the study	In- or outpatient setting, post-acute treatment	
Study design	All types of primary qualitative studies (= no reviews), mixed method studies only if sufficient qualitative data provided (e.g. separate qualitative data analysis)	Single-case studies
Languages	German, English	
Publication status	Published, e-first, full text available	
Type of publication	Journal article or book chapter	

Selection of relevant studies

A PRISMA compliant flow chart [25] will be used to document and summarize the study selection process. Search results will be uploaded to RAYYAN [26], a free software tool allowing reviewer teams to collaborate on systematic reviews online. The selection of relevant studies will be done in groups of at least two reviewers and will comprise the steps listed in table 2. Neither of the review authors will be blind to the journal titles or to the study authors or the institutions.

Table 2: Selection process (Modified after [25])

1	**Objective:**	Building a consented framework of inclusion and exclusion criteria
	Method:	Application and refining of in- and exclusion criteria
	Material:	Titles and abstracts
	▪ A subset of 100 articles out of the search hits will be uploaded to RAYYAN ▪ Each reviewer screens all of the 100 articles. ▪ All reviewers compare their results. ▪ Disagreeing judgements will be discussed and inclusion criteria will be refined through this process. ▪ The consensus process and the modification of inclusion / exclusion criteria (if done) will be documented.	
2	**Objective:**	Consented list of preliminary included studies
	Method:	Screening of all records
	Material:	Abstracts of all articles retrieved

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- All hits are screened and evaluated on the basis of the finally consented criteria.
- The procedure will be done in two pairs of two reviewers (VL&YE, VLen&NP). Each pair evaluates half of the hits.
- The respective pair of reviewers will meet to discuss and compare their lists of preliminary included studies.
- If no consensus is reached, a third reviewer (TM) will read the abstract and moderate the consensus process.
- The two lists of preliminary studies (one list per pair) will be joined to one.
- If modifications of inclusion / exclusion criteria are necessary, this will be done and be documented.

3	**Objective:**	Consented list of finally included studies, list of excluded studies (incl. reason for exclusion)
	Method:	Full-text screening of the preliminary included studies
	Material:	Full text

- All reviewers will read the full text of the preliminary included studies.
- Each reviewer creates a list of finally included and a list of excluded studies (with documentation of reason for exclusion).
- Consensus will be sought regarding in- and exclusion of studies. The process and decisions will be documented.
- The consensus process will result in one list of finally included and one list of excluded studies (with documentation of reason for exclusion).

4	**Reducing number of studies to a meaningful sample**

If the final study sample is too exhaustive to enable a meaningful synthetisation, the number of studies may be reduced by methods described by Benoot et al. [27].

Critical appraisal of study quality

Even though Thomas & Harden [28] describe quality appraisal as a step in their approach to thematic synthesis it has no consequences for study inclusion or exclusion or the weighing of results. Following Noblit & Hare [29] meta-ethnography does not necessarily include quality appraisal of the primary studies. Therefore, we decided not to include explicit quality assessment of the included studies.

Data extraction

For meta-ethnography as well as thematic synthesis pairs of researchers will extract data into spreadsheets independently and discrepancies will be resolved by discussion. Disagreement about extracted data will be resolved by consensus. From studies employing mixed methods designs only qualitative data will be extracted. Extracted background data for each study will include: objective, location, setting, participants and method of data collection and analysis. With respect to the findings, extracted data will include all first and second order constructs on subjective needs and desires related to health care. The distinction of first and second order constructs originated from Schütz' work on the construction of social reality [30]. First order constructs refer to interpretations made by the participants of the primary studies, while second order constructs refer to the interpretation of the study authors. However, the distinction between first and second order constructs are not always clear-cut and meta-synthesis differ in terms of how to deal with original quotes and the inclusion of first order constructs into the analysis [28, 30]. The results of our synthesis are third order constructs [28, 31].

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Choice of the two approaches for synthesizing qualitative data

Out of a variety of possible methods for synthesising qualitative research presented in Booth et al. [14] we chose two diverging ones: Meta-Ethnography (ME) [29] and Thematic Synthesis (ThS) [28]. The decision was based on the attempt to use fundamentally different approaches in terms of epistemology because this impacts subsequent methodological issues. We chose ME as a representative for an idealistic epistemology and ThS representing a realist approach [14].

Description of the two methods for synthesizing qualitative research:

1. Meta-Ethnography

Noblit & Hare developed meta-synthesis as a method to reach a higher level of analysis, than one can expect from single studies and that may lead to new research questions [32]. They defined synthesis as an activity in which separate parts are brought together to form a 'whole' instead of a mere aggregation of data. This construction of the whole is characterized by some degree of innovation, so that the result is greater than the sum of its parts [29].

Meta-ethnography comprises seven phases: In the first phase, the interest of the research has to be determined. Afterwards it has to be decided, which research is relevant to the determined interest (phase 2) and the selected studies will be read (phase 3). This phase is followed by determining, how the studies are related (phase 4). According to Noblit and Hare, three different ways of relation are possible: 1. The studies are roughly similar, 2. the studies are contradictory, and 3. the studies build a line of argument [29]. According to their relation the studies are translated into one another (phase 5). The roughly similar studies will be referred to the reciprocal translation, which means that constructs, metaphors, categories etc. from similar studies are translated into each other. If the studies are contradictory, a refutational synthesis will be done in which contradictory or conflicting results are investigated, aiming to explain the differences between the single studies. [16, 29, 32]. In phase 6 the translations are synthesized. This procedure may result in an overarching system of so-called third-order categories, that means: the interpretation of the reviewer [31]. Additionally, in this phase a line-of-argument synthesis is done. This type of synthesis goes further than translation and permits a broader view on the topic. Phase 7 relies on reflections about where and in which form to publish the results of the synthesis. Noblit & Hare point out that the publishing should be done with respect to the intended audience and the results achieved [29].

2. Thematic Synthesis

Thematic synthesis is a method derived from thematic analysis and was developed by Thomas & Harden in the context of health promotion and public health to address questions about intervention need, appropriateness and acceptability, and factors influencing intervention implementation. This approach shares some characteristics with meta-ethnography, in that the development of descriptive and analytical themes invokes reciprocal translation. Moreover, it has commonalities with grounded theory, in

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that the approach is inductive and themes are developed using a “constant comparison” method. A novel aspect is the recommendation to use computer software to code the results. [28]

The synthesis process consists of three steps: “line-by-line”-coding of the text, developing descriptive themes and generating analytical themes or so-called third order interpretations in the last step [28].

In the first step findings of the primary study will be coded line-by-line. The coding tree develops inductively during the coding process to allow translation of concepts from one study to another. Coding will be done by two team members independently for each study and discrepancies will be discussed in the study team. In step two the descriptive themes are developed. Codes will be organized hierarchically with new codes created to capture the meaning of groups of initial codes (generation of descriptive themes). A synthesis of findings will be generated based on the findings of all original studies and organized by these descriptive themes. During discussion of findings consistency, interrelations between codes and descriptive themes as well as a potential hierarchical order will be examined. The last step is the creation of an interpretative synthesis to go beyond the content of the original studies and create analytical themes. In this stage models or frameworks may be developed to identify and display relationships between the descriptive themes [28].

Application of the two methods

The analysis of the primary studies will be done in two independent working groups that use one of the above outlined methods, respectively.

Discussion

By using two rigorous and codified methods (ME, ThS) for synthesizing qualitative research, we hope to generate insights in needs and desires of people with poststroke aphasia. The identification of all variations of aphasic patients’ needs and desires related to health care may contribute to tailor health care in a more patient-centred way – which is known to improve therapeutic outcomes [33–35]. The combination of two different meta-synthesis methods is unique to our study. To our knowledge such a type of review has never been conducted before and it will allow insights in the implication that methods for synthesizing qualitative research may have on the results of the synthesis.

Comparison of results of the two approaches

The comparison of the results of the two approaches is planned to be done with respect to the content of the categories, the resulting third order constructs, the scope of the results and their potential of generalization and the presumed audience interested in the results (e.g. health-care providers, funders, politicians). As no established method for such a comparison exists and the aspects to consider depend on the results of the respective synthesis, the particular steps of this part will be developed during the procedure and will be described in detail in a forthcoming methodological article.

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This protocol is the result of an interdisciplinary collaborative process involving researchers with different backgrounds and experiences in qualitative research and systematic literature reviews. To limit the potential biases introduced by the conducting researchers, we discuss the design and the results of our research in our interdisciplinary research group for "Qualitative Methods in Healthcare Research" of the German Network of Healthcare Research.

Researcher biases and potential threats to credibility will also be minimized by rigorous and transparent discussion, determination and documentation of all procedures during the review process. Deviations from the original protocol will be documented in PROSPERO.

Ethics and dissemination

Ethical approval is not required. Dissemination of the results will be ensured by publishing the findings of the meta-synthesis in an international peer-reviewed journal. Besides, results will be presented and discussed on national conferences. We will discuss results of the comparison of the different methods of meta-synthesis with the working group for qualitative research of the German Network of Health Care Research (NP and VLen are the spokespersons of the group) and publish a method paper in an international peer-reviewed journal.

Patient and Public Involvement statement: Patients or the public were not involved in the design, or conduct, or reporting, or dissemination plans of our research.

Author Contributions: NP is the guarantor. VLen and NP drafted the manuscript. All authors contributed to the development of the selection criteria, the risk of bias assessment. All authors read, provided feedback and approved the final manuscript.

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Data statement: As this is a study protocol no data is included or available.

Competing interests: None declared.

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ENTREQ-Checklist

		page
Item 1-16		n.a.
17 Coding	Describe the process for coding of data (e.g. line by line coding to search for concepts)	7-8
18 Study comparison	Describe how were comparisons made within and across studies (e.g. subsequent studies were coded into pre-existing concepts, and new concepts were created when deemed necessary).	7-8
19 Derivation of themes	Explain whether the process of deriving the themes or constructs was inductive or deductive	7-8
Item 20-21		n.a.

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PRISMA-P (Preferred Reporting Items for Systematic review and Meta-Analysis Protocols) 2015 checklist: recommended items to address in a systematic review protocol*

Section and topic	Item No	Checklist item	page
ADMINISTRATIVE INFORMATION			
Title:			
Identification	1a	Identify the report as a protocol of a systematic review	1
Update	1b	If the protocol is for an update of a previous systematic review, identify as such	n.a.
Registration	2	If registered, provide the name of the registry (such as PROSPERO) and registration number	2
Authors:			
Contact	3a	Provide name, institutional affiliation, e-mail address of all protocol authors; provide physical mailing address of corresponding author	1
Contributions	3b	Describe contributions of protocol authors and identify the guarantor of the review	9
Amendments	4	If the protocol represents an amendment of a previously completed or published protocol, identify as such and list changes; otherwise, state plan for documenting important protocol amendments	n.a.
Support:			
Sources	5a	Indicate sources of financial or other support for the review	9
Sponsor	5b	Provide name for the review funder and/or sponsor	n.a.
Role of sponsor or funder	5c	Describe roles of funder(s), sponsor(s), and/or institution(s), if any, in developing the protocol	n.a.
INTRODUCTION			
Rationale	6	Describe the rationale for the review in the context of what is already known	2-3
Objectives	7	Provide an explicit statement of the **question(s) the review will address** with reference to participants, interventions, comparators, and outcomes (PICO)	3
METHODS			
Eligibility criteria	8	Specify the study characteristics (such as PICO, study design, setting, time frame) and report characteristics (such as years considered, language, publication status) to be used as criteria for eligibility for the review	4-5
Information sources	9	Describe all intended information sources (such as electronic databases, contact with study authors, trial registers or other grey literature sources) with planned dates of coverage	4
Search strategy	10	Present draft of search strategy to be used for at least one electronic database, including planned limits, such that it could be repeated	4

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Study records:			
Data management	11a	Describe the mechanism(s) that will be used to manage records and data throughout the review	5-6
Selection process	11b	State the process that will be used for selecting studies (such as two independent reviewers) through each phase of the review (that is, screening, eligibility and inclusion in meta-analysis)	5, Table 2
Data collection process	11c	Describe planned method of extracting data from reports (such as piloting forms, done independently, in duplicate), any processes for obtaining and confirming data from investigators	6
Data items	12	List and define all variables for which data will be sought (such as PICO items, funding sources), any pre-planned data assumptions and simplifications	5, table1
Outcomes and prioritization	13	List and define all outcomes for which data will be sought, including prioritization of main and additional outcomes, with rationale	5
Risk of bias in individual studies	14	Describe anticipated methods for assessing risk of bias of individual studies, including whether this will be done at the outcome or study level, or both; state how this information will be used in data synthesis	n.a.
Data synthesis	15a	Describe criteria under which study data will be quantitatively synthesised	n.a.
	15b	If data are appropriate for quantitative synthesis, describe planned summary measures, methods of handling data and methods of combining data from studies, including any planned exploration of consistency (such as I^2, Kendall's τ)	n.a.
	15c	Describe any proposed additional analyses (such as sensitivity or subgroup analyses, meta-regression)	n.a.
	15d	If quantitative synthesis is not appropriate, describe the type of summary planned	7
Meta-bias(es)	16	Specify any planned assessment of meta-bias(es) (such as publication bias across studies, selective reporting within studies)	n.a.
Confidence in cumulative evidence	17	Describe how the strength of the body of evidence will be assessed (such as GRADE)	n.a.

*** It is strongly recommended that this checklist be read in conjunction with the PRISMA-P Explanation and Elaboration (cite when available) for important clarification on the items. Amendments to a review protocol should be tracked and dated. The copyright for PRISMA-P (including checklist) is held by the PRISMA-P Group and is distributed under a Creative Commons Attribution Licence 4.0.**

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Healthcare related needs and desires of older people with post-stroke aphasia: study protocol of a systematic review and qualitative evidence synthesis using two different approaches

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Healthcare related needs and desires of older people with post-stroke aphasia: study protocol of a systematic review and qualitative evidence synthesis using two different approaches

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Keywords: qualitative research, qualitative evidence synthesis, meta-ethnography, thematic synthesis, aphasia

Abstract

Introduction: Stroke is a frequent disease in the elderly population of Western Europe with aphasia as a common consequence. Aphasia is known to impede targeting treatment to individual partients' needs and therefore may reduce treatment success. In Germany, the post-acute care of stroke patients is provided by different health care institutions of different sectors (rehabilitation, nursing, primary care) with substantial difficulties to coordinate services. One goal of our two qualitative evidence syntheses (QES) is exploring distinct health care needs and desires of older people suffering from poststroke aphasia to support the development of integrated care models based on needs of patients who are very restricted to communicate them. Since various methods of QES exist, the aim of the study embedding the two QES, is to determine if results differ according to the approach used.

Methods and analysis: We will conduct two QES by using meta-ethnography (ME) and thematic synthesis (ThS) independently to summarize the results of primary qualitative studies. The main differences between these two methods are the underlying epistemologies [idealism (ME) vs. realism (ThS)] and the type of research question [emerging (ME) vs. fixed (ThS)].

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We will search seven bibliographic databases. Inclusion criteria comprise: patients with post-stroke aphasia, aged 65 and older, studies in German/English, all types of qualitative studies concerning needs and desires related to health care or the health care system. The protocol was registered on PROSPERO, follows PRISMA-P guidelines and includes three items from the ENTREQ checklist.

Ethics and dissemination: Ethical approval is not required. Results will be published in a peer-reviewed journal and presented on national conferences.

PROSPERO registration number: CRD42020171432.

Strengths and limitations of this study

Strengths:

- We will conduct the QES in an interdisciplinary research team with members highly experienced in qualitative research.
- All steps of the research process will be discussed both within our research team and with the interdisciplinary working group "Qualitative Methods in Health Services Research" of the German Network of Health Services Research.
- Study screening and data extraction will be conducted independently in two teams of two authors, and a fifth author mediating any disagreements.

Limitations:

- We will only consider publications in English and German language.

Introduction

Stroke is a frequent disease in Western European countries and the second most common cause of death in Germany and worldwide [1]. However, in Germany, epidemiological data on stroke rely on estimates derived from stroke registries or data from health insurance companies and therefore vary due to methodological differences [2]. Probably 243.000 to 260.000 persons per year are affected by stroke [2], with a clear tendency to risk increasing with age, especially for people aged 65 and older [3]. Aphasia is known to be a very common consequence of stroke: It occurs in about 30% of people with stroke [4]. There are no primary epidemiological data of prevalence and incidence of poststroke aphasia in Germany available. According to Huber et al. [5] in Germany each year about 25.000 people are newly affected by aphasia after stroke. Aphasic symptoms disappear frequently in the short term, but remain in approximately 56% of patients more than ten months after the stroke [6]. This has considerable impact on postacute treatment of stroke with interaction between patient and health care provider being affected by patients' communicative deficits. These deficits are even more important in a highly segregated care system with different logics and payment structures. While we know about the merits of acute care for stroke, little is known about post-acute health care in stroke patients [7]. The coordination of care and the transition between sectors proves to be very challenging [8]. These transitions are prone to overburden patients, especially patients with aphasia, who literally have a problem to raise their voices. There is a substantial need for integrated care and to inform respective health care domains, especially rehabilitation, nursing and primary health care services.

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Several quantitative investigations explored whether stroke survivors' needs are addressed adequately in the health care system [9–15]. All of them revealed a substantial proportion of individuals with unmet needs from at least 51% [13] up to 100 % [12, 16]. These studies clearly show that existing unmet needs are unfavourable related to the severity of disease or disability as well as to psychosocial aspects. For example Andrews et al. [10] showed that unfulfilled needs were associated with greater disability, fatigue and memory problems. Furthermore, data showed a relation to social status, as the risk for having unmet needs was associated with living in deprived areas, low income or being member of an ethnic minority group. Chen et al. [17] discovered, that on average 73.8% of stroke survivors have at least one unmet need. The types of needs mentioned in their systematic review ranged from needs related to body functions, activities and participation or environment to service needs in information, transport, home support / personal care and therapy.

We found three qualitative evidence syntheses (QES) related to our research question [18–20]: Hilari et al. [18] aimed at identifying prognostic factors of health-related quality of life of people with post-stroke aphasia, but not on their needs related to health care. Moreover, they included only three qualitative studies in their narrative synthesis. In their thematic analysis Manning et al. [19] did not focus on health care needs or health care experiences, but were interested more generally in factors contributing to personal recovery and living successfully with aphasia. The meta-ethnography by Pindus et al. [20] focussed on primary and community care and included all stroke patients independent of existing aphasia. Therefore, our review will contribute to answer the question of how health services in Germany could be better suited to the special needs and desires of older people with post-stroke aphasia.

Methods for synthesizing qualitative research: which one to choose when?

Our aim is not only to generate results with respect to the needs of people with poststroke aphasia, but also to contribute to the methodological discussion on QES. Albeit theory-based recommendations for deciding which type of QES to choose for certain qualitative research questions exist [21–23], to date there is no *empirical* evidence, showing in which way and to what extent different methods of synthesizing lead to different challenges and discrepancies during the review process as well as in the findings.

Of the QES methods presented in Booth et al. [21] we chose Meta-Ethnography (ME) [24] and Thematic Synthesis (ThS) [25] because of their ontological positions: ME for epistemological idealism and ThS for epistemological realism. In the spectrum between idealism and realism both methods are closely related [23]: ME as representative of objective idealism**,** that means seeing the world as a result of collectively shared understandings and ThS as a representative of critical realism which assumes, that our knowledge of the 'objective' reality is influenced by our perceptions and beliefs [23, 26]. Although from different points of view – one that assumes there is no reality 'per se' but peoples' shared experience creates reality, and one that assumes there is one reality, but its awareness is violated by our attitudes, abilities, and experiences – the common ground of both may be the idea of people negotiating what reality is.

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With regard to the research question, we chose ME as an approach permitting adjustment of the research question in the course of the review conduction and ThS as an example for a method requiring a fixed research question [21].

Although different in terms of epistemology and research question, both methods are suited to go beyond the findings of primary studies. This is well known to be the case for ME [24, 27, 28], but also confirmed for ThS by Thomas & Harden [25] or Flemming et al. [29], who pointed out that ThS findings may as well generate new interpretations, constructs or hypotheses. The possibility to do so, mainly relies on the quality and depth of the primary studies [29].

Description of the two methods chosen for synthesizing the primary studies

1. Meta-Ethnography

Noblit & Hare developed meta-ethnography as a method to reach a higher level of analysis than one can expect from single studies and that may lead to new research questions [28]. They defined synthesis as an activity in which separate parts are brought together to form a 'whole' instead of a mere aggregation of data. This construction of the whole is characterized by some degree of innovation, so that the result is greater than the sum of its parts [24].

Meta-ethnography comprises seven phases: In the first phase, the interest of the research has to be determined. Afterwards it has to be decided, which research is relevant to the determined interest (phase 2) and the selected studies will be read (phase 3). This phase is followed by determining, how the studies are related (phase 4). According to Noblit and Hare, three different ways of relation are possible: 1. The studies are roughly similar, 2. the studies are contradictory, and 3. the studies build a line of argument [24]. According to their relation the studies are translated into one another (phase 5). The roughly similar studies will be referred to the reciprocal translation, which means that constructs, metaphors, categories etc. from similar studies are translated into each other. If the studies are contradictory, a refutational synthesis will be performed in which contradictory or conflicting results are investigated, aiming to explain the differences between the single studies [23, 24, 28]. In phase 6 the translations are synthesized. This procedure should result in an overarching system of so-called third-order categories, that means: the interpretation of the reviewer [30]. Additionally, in this phase a line-of-argument (LOA) synthesis is done. This type of synthesis goes further than translation and permits a wider view on the topic. Although Noblit and Hare [24] describe the LOA-synthesis as one of three possible methods to be used in step 4, we agree with France et al. 2019 [27] that it should belong to the interpretive synthesis process. Phase 7 relies on reflections about where and in which form to publish the synthesis. Noblit & Hare point out that the publicationshould be prepared according to the intended audience and the results achieved [24].

2. Thematic Synthesis

Thematic synthesis is a method derived from thematic analysis and was developed by Thomas & Harden in the context of health promotion and public health research to

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address questions about intervention need, appropriateness and acceptability, and factors influencing intervention implementation. This approach shares some characteristics with meta-ethnography, e.g., in the reciprocal translation of descriptive and analytical themes. Moreover, there are similarities with grounded theory as the approach is inductive and themes are developed using a "constant comparison" method [25]. The synthesis process consists of three steps: "line-by-line"-coding (1), development of descriptive themes (2) and generating analytical themes (3) in the last step [25].

The first step is coding the whole text of the results' section of the primary study line-by-line and thereby translating concepts from one study to another. In step two new codes are created to capture the meaning of groups of initial codes (generation of descriptive themes). A synthesis of findings will be generated based on the findings of all original studies and organized by these descriptive themes. During discussion of findings' consistency, interrelations between codes and descriptive themes as well as a potential hierarchical order will be examined. The last step is the development of analytical themes, which is seen as an interpretative synthesis that goes beyond the content of the original studies. In this stage models or frameworks may be developed to identify and display relationships between the descriptive themes [25].

Objective

This protocol is part of a broader (comprehensive) research project adressing two questions. First: Which needs and desires in the context of post-acute care do people with post-stroke aphasia have? To answer this question, we undertake a systematic review of the existing qualitative literature using two different aproaches. This part is described here in detail. Beyond that, a second objective of our research is to figure out if, and to what extent, the processes and findings differ according to the respective aproach. The details of the procedures required to answer this question are going to be described in a forthcoming article.

Methods

The protocol was registered under CRD42020171432 with the International Prospective Register of Systematic Reviews (PROSPERO). Changes will be documented in PROSPERO, providing date and rationale for each modification.

The reporting of this protocol is in accordance with the *Preferred Reporting Items for Systematic Reviews and Meta-Analyses Statement for Protocols* (PRISMA-P [31]) and three additional items (item 17-19) from the *Enhancing Transparency in Reporting the synthesis of Qualitative research* (ENTREQ) checklist [32]). The full reports of the reviews will follow the PRISMA [33] and ENTREQ recommendations, the eMERGe guidance (for the meta-ethnography) [34] and partly the EPOC-criteria [35].

Patient and public involvement

No patient involved.

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Search for similar review projects

To verify that our QES are embedded in current research we searched for published and unpublished QES concerning our topic. We searched Medline via PubMed for published and PROSPERO for unpublished QES and found three reviews [18–20] (see introduction section for further details).

PICoS

We chose the PICo tool [36] to formulate our research question. PICo in qualitative research means: **P**opulation, Phenomenon of **I**nterest, **Co**ntext. Our objective is to identify studies that explore health care needs and desires (phenomenon of interest) of people with poststroke aphasia (population) during postacute treatment (context). We used the PICo-scheme complemented by requirements for the **s**tudy design to guide search term selection and inclusion criteria (so-called PICoS-scheme).

Search

Databases and search terms

Search strategies aimed at identifying qualitative research were developed by an experienced Cochrane Information Specialist using an approach guided by relevant literature. First, a preliminary search strategy guided by the PICoS-scheme using text words and controlled vocabulary (Medical Subject Headings) was designed for MEDLINE. Second, twelve relevant articles meeting the inclusion criteria were identified by conducting and screening a scoping search (see Supplementary File 1). These articles were subsequently used to test and refine the MEDLINE strategy. The test revealed that the concept 'stroke' is not consistently mentioned in the abstracts of relevant publications. Therefore, the concept was omitted from the search strategy to achieve the highest possible sensitivity. The resulting search strategy consists of a search block for 'aphasia' and is combined with validated and sensitive search filters to identify qualitative research, where available for the respective database. The refined MEDLINE strategy was adapted to PsycINFO, Psyindex, CINAHL, Web of Science (Social Science Citation Index), ASSIA and a database indexing grey literature: BASE. Supplementary File 1 provides details of all search strategies.

Additional search strategies

In addition to the database search, the review authors will screen the reference lists of included studies.

Inclusion and exclusion criteria

We will use the criteria outlined in table 1 to select appropriate studies. As health care needs differ according to age [37] and most people affected by aphasia after stroke are of advanced age [38], our study focuses on people aged 65 and older.

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Table 1: Inclusion and exclusion criteria for study selection

	Inclusion criteria	Exclusion criteria
Population	Adult People, aged 65 and older, with post-stroke aphasia	Study includes only members of particular (minority) groups
Interest of the study	Needs and desires related to (the) health care (system)	Study focusses special needs related to technical devices Studies examining certain therapeutic means
Context of the study	In- or outpatient setting, post-acute treatment	
Study design	All types of primary qualitative studies (= no reviews), mixed method studies only if sufficient qualitative data provided (e.g. separate qualitative data analysis)	Single-case studies
Languages	German, English	
Publication status	Published or unpublished, e-first, full text available	
Type of publication	Journal article or book chapter	

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Selection of relevant studies

A PRISMA compliant flow chart [33] will be used to document and summarize the study selection process. Search results will be uploaded to RAYYAN [39], a free software tool allowing reviewer teams to collaborate on screening references online. The selection of relevant studies will be performed in groups of at least two reviewers and will comprise the steps listed in table 2: Parallel to the identification of content-related in- and exclusion criteria, method-specific in- and exclusion criteria will be gathered from the methodological literature (step 1). Content-related criteria are adopted to screen the search results and establish a list of preliminary included studies (step 2). The full-text of the studies will be read in order to remove studies not fulfilling the content-related inclusion criteria in detail (step 3). The resulting sample from this step will be examined using the method-specific criteria (step 4). Neither of the review authors will be blind to the journal titles or to the study authors or the institutions.

Table 2: Selection process (Modified according to [33])

1	**Objective:**	Building a consensus-based framework of content-related inclusion and exclusion criteria and gathering method-specific criteria
	Method:	Application and refining of content-related in- and exclusion criteria, formulation of method-specific in- and exclusion criteria
	Material:	Titles and abstracts of a random sample of 100 articles for content-related and methodological literature for method-specific criteria

- A subset of 100 articles randomly selected out of the search hits, will be uploaded to RAYYAN
- Each reviewer screens all of the 100 articles.
- All reviewers compare their results.

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- Disagreeing judgements will be discussed and inclusion criteria will be refined through this process.
- Pairs of researchers (VLev&YE, VLen&NP) derive method-specific in- and exclusion criteria from the methodological literature.
- The consensus process and the modification of content-related and method-specific inclusion / exclusion criteria (if done) will be documented.

2	**Objective:**	Consensus-based list of preliminary included studies based on content-related criteria
	Method:	Screening of all records using content-related in- and exclusion criteria
	Material:	Abstracts of all articles retrieved

- All hits are screened and evaluated on the basis of the finally consented content-related criteria.
- The procedure will be done in two pairs of two reviewers (VLev&YE, VLen&NP). Each pair evaluates half of the hits.
- The respective pair of reviewers will meet to discuss and compare their lists of preliminary included studies.
- If no consensus is reached, a third reviewer (TM) will read the abstract and moderate the consensus process.
- The two lists of preliminary studies (one list per pair) will be joined to one.
- If modifications of content-related inclusion / exclusion criteria are necessary, this will be done and be documented.

3	**Objective:**	Consensus-based list of finally included studies based on content-related criteria, list of excluded studies (incl. reason for exclusion)
	Method:	Full-text screening of the preliminary included studies using content-related in- and exclusion criteria
	Material:	Full text

- All reviewers will read the full text of the preliminary included studies.
- Each reviewer creates a list of finally included and a list of excluded studies (with documentation of reason for exclusion).
- Consensus will be sought regarding in- and exclusion of studies. The process and decisions will be documented.
- The consensus process will result in one list of finally included and one list of excluded studies (with documentation of reason for exclusion).

4	**Objective:**	Method-specific, consensus-based lists of finally included studies, lists of excluded studies for each method (incl. reason for exclusion)
	Method:	Full-text screening of the studies included in step 3 using method-specific in- and exclusion criteria
	Material:	Full text

- Method-specific review teams will read the full text of the studies included in step 3
- Each reviewer creates a list of studies to be included on the basis of method-specific criteria (with documentation of reason for exclusion).
- Consensus about in- and exclusion will be sought in the method-specific teams. The process and decisions will be documented.
- This process will result in two lists of studies to be in- or excluded

Critical appraisal of study quality

Noblit & Hare [24] did not take a stance on quality appraisal of the primary studies included in meta-ethnography. Other authors of meta-ethnographic methodological research present contrasting opinions on quality appraisal: Whereas Atkins et al. [28], Campbell et al. [40], Froud et al. [41], Pound et al. [42] or Toye et al. [43] seem to be willing to explore the methodological quality of studies to be included in meta-ethnographies, Britten et al. [30] or France et al. [27] point out, that time could be spent more efficiently by concentrating on the topic of the review [30] or that critical appraisal is redundant as long as it has no implications for in- or exclusion of studies [27]. Thomas & Harden performed critical appraisal of the primary studies included in their thematic synthesis and undertook a sensitivity analysis to determine the impact of low-quality studies on the findings of the review [25]. Overall, the position of qualitative researches

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on this topic is still indecisive. What convinced us to include critical appraisal, were the findings of Pindus et al. that a good quality score does not necessarily mean, that the study provides sufficient contextual data [20], and France et al. [27], that poor descriptions of methods do not necessarily equate to a poorly-conducted study. Toye et al. [44] drew attention to the fact, that insightfulness and methodological rigour are different facets of quality and may be quite different. So, we felt that the determination of methodological quality could contribute to the ongoing debate, as long as we go beyond the mere exclusion of methodologically flawed studies [44, 45]. We will follow Thomas & Harden [25] and their sensitivity analysis approach and further will explore the relation between insightfulness and methodological quality in the included studies. To determine the methodological quality of the included studies we chose the widely used Critical Appraisal Skills Programme (CASP) Checklist for qualitative studies [46–48].

Data extraction and analysis

For meta-ethnography as well as for thematic synthesis researcher pairs will independently extract data into spreadsheets and resolve potential discrepancies by discussion. Unsolved disagreements will be discussed with a fifth researcher. From studies employing mixed methods designs only qualitative data will be extracted. Extracted background data for each study include: study-design, objective, location, setting, participants and method of data collection and analysis.

For the meta-ethnographic analysis second order constructs on subjective needs and desires related to health care will be extracted. Second order constructs are defined as the interpretation of the authors of the primary study [43]. First order constructs (the interpretations of the participants of the primary studies) are to be additionally included in our analysis only if clarification of the second order constructs is required [27]. The second order constructs will lead to third order constructs, in other words: the interpretation of the reviewer team [25, 30]. For the thematic synthesis all of the text in the 'results' section is to be analyzed in three steps: First, a line-by-line coding, secondly the organization of the resulting codes to descriptive themes and thirdly the development of analytical themes out of the descriptive ones. These analytical themes correspond to the third order constructs of the meta-ethnographical analysis [25] and therefore can be used to compare the findings of ME and ThS.

Constant comparison process

Two independent working groups will each conduct a review using one of the outlined methods. Beyond that, our intention is to compare the challenges, questions, and results arising from each step of the review process to generate empirical material that may clarify the similarities as well as the differences of the two methods. For there is no existing framework that could guide us through the process, we will develop the distinct steps of the comparison during the review process using the method of constant comparison. As an example, we describe how we will proceed if the selection process ends up in two different study samples: To avoid that further differences, particularly the findings of the synthesis in terms of third order constructs / analytical themes, are seen as a result of the divergent study samples, we will analyse both samples with each

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method. That means: First each group analyses the studies fulfilling the content-related *and* the specific inclusion criteria of 'their' method, and afterwards they include the method-specific studies of the other group – an approach very similar to sensitivity analysis suggested by Thomas & Harden [25]. This will enable us to distinguish between differences resulting from the methodological steps applied and differences based only on the fact, that different studies were included in the QES.

The findings of the two approaches are to be compared with respect to the process of conduction, the resulting third order constructs (ME) / analytical themes (ThS), the scope of the findings and their potential of generalization. Because we expect challenges to occur while performing the particular steps of the review, we decided to determine the exact procedures during the process – a strategy named "Gegenstandsangemessenheit" and seen as an indicator of quality in qualitative research [49–51]. To make this process visible, we will document the respective problem, our discussion and the way we decide to solve it – including the reason for the decision - in a forthcoming methodological article.

A systematic comparison between two qualitative synthesis approaches cannot limit itself to be descriptive, but will have to be interpretive to explain the differences. The point of departure of this analysis will be a comparison of the findings of both approaches. Differences and similarities will be identified. In a second step we will try to understand the differences by looking closer at the respective methods, processes and decisions made during the conduct of the respective study. This will enable us to identify reasons for decisions made and to distinguish between possible reasons that are grounded in the respective methodology or approach vs. other reasons, such as personal backgrounds of the reviewers or special theoretical stances. By doing this we make reference to the comprehensive meta-study approach and plan to explain differences between the findings of the two approaches due to reasons related to data analysis, methods and theories applied [52].

Author background and reflexivity

All participating reviewers are experienced qualitative researchers, with different clinical and scientific backgrounds: psychology / health services research, lecturer for qualitative research methods, papers on e.g. qualitative data acquisition and analysis (NP); psychology / rehabilitation sciences and health services research; lecturer in interpretative qualitative methods, papers on e.g. quality criteria of qualitative research and qualitative sampling (TM); gerontology / palliative care (VLev, YE) and Public Health / nursing / physical therapy, lecturer for qualitative and health services research (VLen). This will enable us to adopt a comprehensive view on the topic investigated. Constant discussion and regular meetings will help to avoid skewing the analysis and interpretation and thus minimize reviewer bias. Potential threats to credibility as well as reviewer bias will also be minimized by rigorous and transparent discussion in our interdisciplinary research group "Qualitative Methods in Health Services Research" of the German Network of Health-Services Research, as well as determination and documentation of all procedures during the review process. To disclose how our positions may have influenced the review and whether our attitudes toward the research topic changed during the review, we will include a reflexive statement [35] in the results section of our review.

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Discussion

By using two rigorous and codified methods (ME, ThS) for synthesizing qualitative research, we expect to generate in-depth conceptual understanding of what is important for people with poststroke aphasia. The identification of all variations of patients' (living with aphasia) needs and desires related to health care may contribute to tailor health care in a more patient-centred way – which is known to improve therapeutic outcomes [53–55]. The combination of two different QES methods is unique to our study. As far as we know such type of review has never been conducted before and it will allow insights on the impact the QES methods may have not only on the findings but on the whole review process.

A limitation of our study is that we will not include publications in languages other than German or English. This might limit the range of the results of our QES.

In addition, a comparison of different QES methods can only be made with some methodological limitations. From a pure conceptual or even idealistic perspective, a strict comparison is hardly possible or only possible for two quite similar approaches. We chose two approaches similar enough to answer the same research question (leaving it open to adopt the question during the course of the study if necessary). We assume, however, that the reality of choice over QES approaches is in practice much more dependent on criteria that go beyond methodological considerations – such as personal background and familiarity with a method, personal preferences of team members etc. [21]. In practice, therefore, boundaries between different QES approaches might be blurred with regard to concepts, methods or terms. Since there is no *ideal* comparison in this vein, we find it appropriate to make a few restrictions to the respective QES methodologies in order to allow for substantive comparisons. It is well acknowledged that these restrictions should be an explicit part of the analysis, i.e. the approach to comparison.

Ethics and dissemination

Ethical approval is not required, because no human subjects or personal data are directly involved in our review. To ensure dissemination, the findings of the QES will be published in an international peer-reviewed journal and presented on national conferences. A distinct method paper will be published in an international peer-reviewed journal.

Authors' contributions: NP is the guarantor. VLen and NP drafted the manuscript. NP, VLen, YE, VLev and TM designed the study. MIM designed the search strategy. All authors read, provided feedback on and approved the final manuscript.

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Competing interests: None declared.

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Supplementary File 1: Search strategies and relevant references used to test the MEDLINE strategy

Medline (PubMed)

(comprises all terms of "HIRU hedges Qualitative – MEDLINE filter, best balance of sensitivity and specificity" and MeSH and textwords identified during text analysis of twelve relevant references, adding sensitivity)

((Aphasia[mh] OR aphasia*[tw]) AND (Qualitative Research[mh] OR Interviews as Topic[mh] OR Interview[pt] OR Focus Groups[mh] OR experience*[tw] OR qualitative[tw] OR interview*[tw] OR needs[tw] OR themes[tw] OR perspective*[tw]))

PsycINFO (Ovid SP)

1. Aphasia/
2. aphasia*.tw.
3. 1 or 2

[HIRU hedges Qualitative – PsycINFO filter, best balance of sensitivity and specificity]

4. experiences.tw. or interview*.tw. or qualitative*.tw.
5. 3 and 4

Psyndex (PubPsych)

(aphasi* AND (experience* OR qualitativ* OR interview* OR needs OR themes OR perspective*)) AND DB="PSYNDEX"

CINAHL (EBSCO)

1. MH "aphasia+"
2. TI (aphasia*) OR AB (aphasia*)
3. S1 OR S2

[Wilczynski 2007 "small drop sensitivity"-filter]

4. MH "attitude+" OR MH "interviews+" OR MH "qualitative studies"
5. S3 AND S4
6. MH "Interview+" OR MH "Grounded theory" OR MH "Focus Groups" OR MH "Field Studies" OR MH "Life Experiences+"
7. S3 AND S6
8. S5 OR S7

Wilczynski NL, Marks S, Haynes RB. Search strategies for identifying qualitative studies in CINAHL. Qual Health Res. 2007 May;17(5):705-10.

Social Science Citation Index (Web of Science)

((TS=aphasia*) AND (TS=qualitative OR TS=interview*))

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ASSIA (ProQuest)

ALL(aphasia*) AND ALL(qualitative OR interview* OR focus group* OR experience* OR need* OR theme* OR perspective* OR attitude*)

Bielefeld Academic Search Engine (BASE – Advanced Search)

[limited to doctype = Dissertationen und postgraduale Arbeiten]

(aphasia*) AND (qualitative OR interview* OR experience* OR needs OR themes OR perspective*) doctype:183

Twelve relevant references used to test the MEDLINE search strategy

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Reviewer: 1
Reviewer Name
Jane Noyes
Institution and Country
Bangor University and Cochrane
UK
Reviewer(s)' Comments to Author:
Please state any competing interests or state 'None declared': None declared

Commentary Reviewer 1	Answer	Manuscript Page
1. If conducting the work as a methodological exemplar, this this should be an explicit aim and there should be an additional methodological question to address. Please address	Dear Mrs. Noyes, thank you very much for your time and efforts spent on reviewing our manuscript. Your comments were very helpful to us. The primary aim of the manuscript is to function as a study protocol for two meta-syntheses which are aimed at answering a specific research question. The secondary aim of our project is to compare the process and results of the two meta-synthesis methods. This topic will be addressed in further publications. We appreciate your interest in the method of comparison, but the development of the methodology is part of our future study. Right now we would prefer our manuscript to be viewed as a study protocol for two seperate meta-syntheses. Either way, we will give an outlook of the route we will travel in comparing both methods. Further refinements of this approach will be described in later methodological publications. To clarify this, we reformulated the research question, please see under "Objectives" in the introduction.	5
1.1 The title needs looking at again. This is not a double meta-synthesis. It is a methodological exemplar comparing the application of two qualitative evidence synthesis methods. Please address.	As already mentioned, our aim is twofold: Gathering the evidence of qualitative studies on a certain topic (which is described here in detail) AND *later on* examining the differences (and similarities) of the different methods. The term "double-metasynthesis" may be misleading, therefore we changed the title. New Title: Healthcare related needs and desires of people with post-stroke aphasia: study protocol of a systematic review and qualitative evidence synthesis using two different approaches	1
1.2 Of particular concern, the framework to compare and contrast the respective findings from the syntheses will be described in detail elsewhere. This means that the current protocol is incomplete and the main purpose cannot be achieved. Please address by including the methods for developing the structured framework to compare the two methods.	Please see the new "Objectives" section and "Constant comparison process" in the methods section.	5,9
Otherwise this protocol is largely redundant in purpose.	Before starting our research, we conducted a search for existing qualitative synthesis in order to make sure, our research is NOT redundant. This revealed, that our synthesis was the only one focussing on needs and desires of older people with post-stroke aphasia with respect to the health care system. We described the differences between this three reviews and ours in the introduction.	3

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Commentary Reviewer 1	Answer	Manuscript Page
I acknowledge that this framework is likely to evolve during the conduct of the reviews, but the protocol **at least should outline the author's theoretical conceptualisation and ideas on this and articulate the processes looking forward**. This is the fundamental methodological component of this protocol. Otherwise its just a protocol for the conduct of the same review using two reasonably similar methods	Please see section "data extraction and analysis" and "Constant comparison process"	9
1.2.1 Thus far methodological exemplars have compared methods that start at different positions and process the data in completely different ways to produce different types of synthetic product, such as framework or best fit framework synthesis compared with thematic synthesis. **The manuscript could be further strengthened by further articulating the epistemological differences between the two selected methods and how the team plan to 'operationalise' their thinking and processing of these epistemological differences in processing the data in the respective reviews.** This issue is particularly important as both selected methods have more similarities than differences in their anticipated purpose and data processing – that is both methods are designed to produce new theory and both methods share similar aspects of data processing and synthesis so the main key difference is the epistemological location.	Please see "Methods for synthesizing qualitative research: which one to choose when?"	3
2. The authors appear to misinterpret the current guidance on choosing a qualitative evidence synthesis methodology. 'Albeit theory-based recommendations for deciding which type of synthesis to choose for certain qualitative research questions exist [14–16], to date there is no empirical evidence showing which approach serves best in qualitative health care research.' This sentence demonstrates a lack of understanding of the published guidance in that the choice of qualitative evidence synthesis method is dependent on a number of methodological considerations (such as the question, number and type of studies, richness and quality of the data etc). The RETREAT framework has been specifically designed to guide authors to select the most appropriate qualitative evidence synthesis method for their specific healthcare context. Please reread: https://www.ncbi.nlm.nih.gov/pubmed/29548841 and https://www.integrate-hta.eu/wp-content/uploads/2016/02/Guidance-on-choosing-qualitative-evidence-synthesis-methods-for-use-in-HTA-of-complex-interventions.pdf	We referred to a lack of empirical (sic!) evidence in the comparison of different QES approaches. The valuable published RETREAT guidance is based on a review of experiences and deliberations of authors regarding different potentially relevant dimensions that should be acknowledged in the decision for or against certain QES approaches. It is not a substitute for an empirical comparison, which our projects aims at. We used the RETREAT framework to get an overview over existing QES-methods. We have tried to make this more explicit in the introduction under "Methods for synthesizing qualitative research: which one to choose when?"	3

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Commentary Reviewer 1	Answer	Manuscript Page
2.1 The key point here is that although the methods for the search strategy would be the same irrespective of the method of synthesis, you may select a different qualitative evidence synthesis method depending on the type and richness of available studies and you may sample the studies differently depending on the qualitative evidence synthesis method selected. A meta-ethnography question tends to be constructed differently than a thematic synthesis question (for example). In the current protocol, all the methods of question formulation, data identification, selection and processing are generic up until the point of synthesis and this would not happen if the syntheses were conducted as different and separate products. It needs acknowledging that specific processes would not happen in this way if the two syntheses were conducted separately. This may also lead to a distortion of the methodological exemplar and its findings.	We are quite aware that a comparison of different QES methods can only be made with some methodological limitations. From a pure conceptual or even idealistic perspective, a strict comparison is hardly possible or only possible for two quite similar approaches. We have decided to choose approaches similar enough to be able to apply to the same research question (leaving it open to adopt the question during the course of the study if necessary), and to base (!) it upon the same set of studies, which does not predeterime the final selection of studies or the priorities given to each study. We assume, however, that the reality of choice over QES approaches is in practice much more dependent on criteria that go beyond methodological considerations – such as personal background and familiarity with a method, personal preferences of team members etc. In practice, therefore, boundaries between different QES approaches might be quite blurred in terms of concepts, methods or terms. Since there is no "ideal" comparison in this vein, we find it appropriate to make a few restrictions to the respective QES methodologies in order to allow for a substantive comparions. It is well acknowledged that these restrictions should be an explicit part of the approach to comparison.	11
By using PRISMA P which is designed for quant SRs and meta-analyses- the authors have been taken down the path of writing a protocol as if all the methodological uncertainties were known. For example, current Cochrane guidance suggests that final selection of a method of qualitative evidence synthesis may not be appropriate until the pool of available evidence is known and an assessment of its methodological quality and conceptual richness can be made. The authors may want to look at the **Cochrane template for undertaking a qualitative evidence synthesis** (see link below). Not all aspects will be relevant (such as integrating the QES with a Cochrane intervention effect review, but other issues not covered in the current protocol such as **author reflexivity should be included in the current protocol**. See: https://epoc.cochrane.org/sites/epoc.cochrane.org/files/public/uploads/Resources-for-authors2017/epoc_qes_protocol_and_review_template.pdf	The EPOC-guidance offers very useful aspects to consider when writing an abstract or a plain language summary. The aspects the EPOC-guidance recommends for writing abstracts are fulfilled. For our paper is a protocol and does not belong to the systematic of Cochrane Reviews, we did not search for guidance in this area. Nevertheless we decided to expand our protocol with a statement on author reflexivity, which is suggested to be embedded in plain language summaries. Please see "Author background and reflexivity" in the method's section. The final review will incorporate a reflexive statement to disclose how our positons may have influenced the review and whether our attitudes toward the research topic changed during the review. Please see "Author background and reflexivity" in the method's section.	10 10
The current protocol will only therefore 'work' if it is identified as a methodological exemplar and it is acknowledged that if conducted separately then the reviews would likely look different. For example, in the current protocol the assumption is that the review team want to use the same evidence (ie studies) to conduct both types of synthesis. In reality, if conducted separately different types of evidence may be selected for the respective qualitative evidence synthesis method. Please address.	Please see new text under „Objectives" and „Constant comparison process".	5, 9

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Commentary Reviewer 1	Answer	Manuscript Page
2.2 Linked to 2.1 **The question is exceptionally broad** "What are the needs and desires related to health care of people suffering from aphasia after stroke?" This is not a great QES question as it is so broad and could cover just about anything for any group of people with stroke and aphasia in any context or health system. A more specific and meatier question could enable a much richer and more meaningful methodological exemplar.	Broad questions in qualitative synthesis are not exceptional, because in the beginning it is unclear which – if any – qualitative evidence does exist. "The exploratory process means that initial qualitative review questions may be broad in order to map what is known, before formulating or refining questions" (Harris et al. 2018, p 4). Moreover, we do not see, why our question should be too broad, while questions labelled as "useful" in Booth et al. 2018, p. 48-49 seem to be quite similar to ours: Question 1. "To analyze, through an interpretative systematic review of qualitative studies, the meaning and motivation of the WTHD in patients with chronic illness or advanced disease" Question 2: "What are the experiences of adolescents and young adults living with everyday pain?" We could not discover the fundamental difference to our research question ("What are the needs and desires related to health care of people suffering from aphasia after stroke?") To ensure that there are at least some qualitative studies in this area, we conducted searches in advance (for primary studies as well as for reviews), that resulted in some, but not exhaustive evidence. By studying similar – not identical – reviews found, we realized that the corpus of the existing evidence is not very large. The three qualitative synthesis focusing on questions related to ours included 51 (Pindus et al. 2018), 31 (Manning et al. 2019) and 3 studies (Hilari et al. 2012). Only Hilari et al. (2012) and Manning et al. (2019) restricted their research on stroke survivors with aphasia (but not on older people). Pindus et al. included subjects with and without aphasia. Therefore, to formulate a narrower question would probably result in very few hits in the database-search or after application of in- and exclusion criteria.	3
Plus, even though it is very broad, as specified, the question is more suited to a thematic synthesis whereby you might include conceptually thin clinical study reports of relatively small scale qualitative studies.	Conceptually "thin" studies are not the aim of our search, albeit they do not violate research findings. They just do not add much to the synthesis findings. Using the concepts of "needs and desires" opens up the possibility to use both a rather descriptive approach – such as the thematic analysis –and a more interpretative approach, such as the meta-ethnography approach. Therefore, we do not really comply with the point raised here.	
For a meta-ethnography you need at least some conceptually rich qualitative studies in order to develop new theory (the main purpose of conducting a meta-ethnography). Although both meta-ethnography and Thomas and Harden's thematic synthesis method have the capacity to develop new theory or analytical level interpretations not seen in primary studies**, it is not clear what added value a new theory of stroke survivor's needs and desires from healthcare would add to what is already known. In summary, the question as currently specified seems the least likely to offer the most opportunities to draw out the differences in the findings from the two methods, when the** differences are likely to be at theoretical level. Please address	The point of departure for our research question comes from the problems we see in the German health care system with regard to cross-sectoral care. While the different health care institutions caring for stroke patients all have their merits, there is a huge problem in coordination of care and taking responsility of care, which is often left to patients overwhelmed with their situation. Especially patients with aphasia have a literal problem to raise their voice. Therefore, we want to learn from qualitative studies about the needs and desires (i.e. objective needs as reported by the patients themselves, as well as subjective needs related to goals, preferences, expectations etc.) in order to inform present health care approaches that aim to integrate the different care stakeholders, especially rehabilitation, nursing and primary helath care services. From our view, we find it difficult to predict the added value(s) of our analyses. At least, our review will provide a systematic overview on "what is known", and depending on the original studys and the type of analysis also what can be added to the knowledge of the primary studies.	
2.2.1 The 'I' in PICO relates to the 'phenomenon of Interest'. Please make this clear in the protocol	„phenomenon of" added, see PICoS	6

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Commentary Reviewer 1	Answer	Manuscript Page
3. Please refer to 'finding's and not 'results'. This is a qualitative exemplar so please use the language most relevant to the qualitative paradigm.	As our synthesis will lead to different types of results (related to the question of needs and desires of people with post-stroke aphasia (1) and related to methodology (2)) the use of the term "finding" in the sense Lewin et al. (2015) defined it:*"For the purposes of CERQual, we define a review finding as an analytic output from a qualitative evidence synthesis that, based on data from primary studies, describes a phenomenon or an aspect of a phenomenon."* Refers only to the first objective. But we acknowledge that in the paragraph „Constant comparison process", the term "finding" is more suitable and revised the text.	9
4. The lack of critical appraisal is problematic for the reasons already outlined. The review authors will need to make an assessment of methodological limitations in included studies (such as thick thin data and conceptually rich poor data) in order to understand and compare the methodological processing of data extracted from included studies. If sampling, the author team may also want to sample based on an assessment of methodological limitations in primary studies. Whilst I accept that there is ongoing discussion about the appropriateness of current appraisal tools, and whilst we await the release of the new CAMELOT tool, Cochrane has however provided guidance on a narrow set of quality appraisal tools that are appropriate for use in qualitative evidence syntheses if they are used to identify methodological limitations.	Rose Barbours' paper from 2001 in the BMJ pinpointed the problem of using quality appraisal tools for so-called qualitative studies. As we have analysed in a paper of the qualitative methods working group within the German network on Health Services Research (Stamer et al. 2015), there is a substantial problem in a general appraisal of methodologically different studies, although they firm under the heading of being "qualitative". To our knowledge the classification of data as "thin" or "thick" in qualitative synthesis is not necessarily a result of good methodological quality but of the insightfulness of the data (Toye et al. 2013). Pindus et al. (2018), who did quality appraisal, found that "while the included studies achieved good quality scores, many failed to provide sufficient contextual detail". Despite our reservations against critical appraisal, which we share with many others (e.g. Garside et al., 2014), we are quite aware that there is always some sort of appraisal involved when we read and analyse qualitative studies, although it is very challenging to come up with operational criteria that work for different qualitative study types. Still, we have decided to use the CASP-Tool to critically appraise the primary studies included in the review. Please see "Critical appraisal of study quality" in the method's section.	8
5. Application of GRADE CERQual may provide another opportunity for the authors to further articulate the differences in the confidence in review findings from the respective application of QES methods.	Thank you for this hint to the GRADE-CerQual – Tool. We thought about using it, but decided to rely on the longer established ENTREQ tool. "… the 'Enhancing Transparency in the Reporting of Syntheses of Qualitative Research' tool (…) was the first reporting tool for QES and consists of 21 items within the five overarching categories. As a generic tool it documents the most frequently used methods for QES to which it might apply, acknowledging that the approaches and methodology for synthesis are usually driven by the research question posed" (Flemming et al. 2019, p 8) Moreover our interest laid in using a a widely accepted guidance for a protocol because our experience in publishing qualitative research showed so far, that publishers (and reviewers) very often refer to the "classical" quantitative paradigm.Ttherefore we decided to combine PRISMA-P, a checklist, that "is the most widely applicable across ***different*** research areas" (Siddaway et al. 2019, p 9.14) with ENTREQ. We added information on use of EPOC for the final review on page 5.	5

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Commentary Reviewer 1	Answer	Manuscript Page
6. The author team have not cited the eMERGe guidance on reporting meta-ethnography or the latest methodological articles on articulating and operationalising the complex analytical phases of meta-ethnography.	The eMERGe guidance was planned to be used as guidance for the description of the analytical steps of the meta-ethnography in the main paper.	5
7. Of particular note, both the Thomas and Harden method of thematic synthesis and meta-ethnography have commonly been poorly .. applied and not conducted as intended. It is common to see published reviews that 'use' these methods but various aspects have been adapted, skipped over or misunderstood. To produce a convincing methodological exemplar, the authors will need to stay faithful to the methods as originally intended and follow the guidance of the originators or authors whose methodological work to further articulate the methods is considered high quality by the originators. Thank you once again for asking me to review this interesting protocol. Overall, comprehensive revision and additional thinking is required.	We very much agree with the point and will try to make appropriate references to the original works. Still, the essence of qualitative methods is what in German we call "Gegenstandsangemessenheit" (appropriateness with regard to the object of study). We are always called upon to reflect on the appropriateness of the methods and be transparent in our methods or methodological choices and to change them when necessary. Rigour should not be confused with methodological orthodoxy. This characteristic of qualitative research relates both to orginal research and QES. Meta-Ethnography is a very good example to show that the origin is not designed as a step by step practice framework (even the authors mention this). Many others (Atkins, Britten, Cahill, Campbell, France, Toye …) refined the methodological steps of meta-ethnography but it is definitely compatible with the epistemology of qualitative research, that the "only and right one" does ***not*** exist. Of more importance in qualitative research is transparency and detailed description of the application of the respective method.	

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Reviewer: 2
Reviewer Name
Assoc Prof Mandy Stanley
Institution and Country
Edith Cowan University, Australia
Please state any competing interests or state 'None declared':
None declared.

Please leave your comments for the authors below

Commentary Reviewer 2	Answer	Manuscript Page
Thank you for the opportunity to review the manuscript. I am intrigued by the ideas of analysing the data using two different approaches to synthesis and look forward to reading the outcome. It is indeed an interesting question. However by naming it a "double meta-synthesis" seems to me to be misleading. Double synthesis implies to me that you are analysing the data twice over, and whilst essentially you are, it is more that you are approaching the same data in two different ways. It may be a matter of semantics but I think it is important to reflect on	Dear Prof. Stanley, first of all we would like to thank you for your helpful comments on our manuscript. The title has been changed to "Healthcare related needs and desires of older people with post-stroke aphasia: study protocol of a systematic review and qualitative evidence synthesis using two different approaches".	1
You say you will search at least four databases. Will it be four or more than four?	Please see new description under "databases and search terms" in the method's section.	6
There appears to be some material out of place in the ethics and dissemination section of the abstract.	Thank you for this hint, we revised the text.	
Will you inform policy? How? Much more likely to inform best practice.	The point of departure for our research question comes from the problems we see in the German health care system with regard to cross-sectoral care. While the different health care institutions caring for stroke patients all have their merits, there is a huge problem in coordination of care and taking responsility of care, which is often left to patients overwhelmed with their situation. Especially patient with aphasia have a literal problem to raise their voice. Therefore, we want to learn from qualitative studies on the needs and desires (i.e. objective needs as reported by the patients themselves, as well as subjective needs related to goals, preferences, expectations etc.) in order to inform present health care approaches that aim to integrate the different care stakeholders, especially rehabilitation, nursing and primary helath care services.	2
I was interested in the decision to focus on older people (rather than elderly) yet you cite statistics about the frequency of stroke from age 40-79.	We added some new research more focused on the age group 65 years and older (see Introduction).	2
The manuscript needs review for a number of missing words and some smoothing out of English language.	The manuscript has been revised.	
I got to the end of the introduction section and was unclear how the findings about needs related to people with aphasia, and why the review needed to be conducted. This needs to be much more explicit, not just the methodological curiosity in trying two different analysis approaches.	Please see the revised introduction.	3
In the methods section there is a statement about ethics which is not necessary here as stated later.	We placed the ethics statement at the end of the paper.	11

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Commentary Reviewer 2	Answer	Manuscript Page
Databases and search terms: state the languages to be searched in	We did not restrict the search to certain languages.	
Table 2 objective Is it consented or consensus?	Text changed to consensus-based.	Table 2
Data extraction - will you also extract study design?	We added "Study design" (see "Data extraction and analysis")	9
You talk about the distinction between first and second order constructs not being clear, what approach will you use?	Because it may confuse readers, we decided to remove the sentence about this problem.	
The section on how you chose the two different approaches needs to come earlier to make a bit more sense.	This part was modified and shifted to the introduction.	3
Am I correct in understanding that the analysis will be done by two independent teams? I was wondering how you would prevent the findings from one influencing the other especially if they are both discussed with the research network.	Discussing findings with other researchers, so called triangulation, is a common procedure in qualitative research and aims to prevent scientists from overlooking further interpretations. One may call this influencing each other, but in qualitative research this is deemed to be adequate because it may enhance the validity of the findings.	
The discussion is a bit weak. Aphasic patients should be people living with aphasia. You suggest it will inform more patient-centred ways but how do you know current ways are not? The findings should provide a more in-depth conceptual understanding of what is important for people living with aphasia..	We revised the term aphasic people to people living eith aphasia. Studies supporting our view that current ways are not adequately patient-centered (see first paragraph on p.3): "Several quantitative investigations explored whether stroke survivor's needs are addressed adequately in the health care system [5–11]. All of them revealed a substantial proportion of individuals with unmet needs from at least 51% [9] up to 100 % [8, 12]. These studies clearly show that existing unmet needs are related to the severity of disease or disability as well as psychosocial aspects, living in deprived areas, low income or being member of an ethnic minority group."	3
I am also intrigued about how you will manage reporting. If you publish one approach will that make publication of the other approach more difficult? It clearly will lead to an interesting methodological paper however that shouldn't be the primary goal	We plan to publish both reviews and an additional one about the methodological challenges and how we solved them.	

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Reviewer: 3
Reviewer Name
Betje Schwarz

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German Pension Insurance Berlin-Brandenburg,
Germany

Please state any competing interests or state 'None declared':
None declared.

Please leave your comments for the authors below

Kommentar Reviewerin 3	1. Antwortvorschlag / Anmerkung 2. Formulierungsvorschlag neuer Text	Ggf. Quellenangabe
A very good study protocol of a very interesting and promising study	Dear Mrs. Schwarz, thank for your praise of our manuscript and the efforts spend on reviewing it.	
I have only one request for clarification: in the section 'Authors Contributions' you state, that "All authors contributed to […] the risk of bias assessment." Which bias assessment?	Thank you for this hint. The sentence related to risk assessment was printed unintentionally and now has been deleted.	
You also state, that you will not assess the methodological quality of the primary studies: "Even though Thomas & Harden [28] describe quality appraisal as a step in their approach to thematic synthesis it has no consequences for study inclusion or exclusion or the weighing of results. Following Noblit & Hare [29] meta-ethnography does not necessarily include quality appraisal of the primary studies. Therefore, we decided not to include explicit quality assessment of the included studies."	We revised the part on critical appraisal.	8

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PRISMA-P (Preferred Reporting Items for Systematic review and Meta-Analysis Protocols) 2015 checklist: recommended items to address in a systematic review protocol*

Section and topic	Item No	Checklist item	page
ADMINISTRATIVE INFORMATION			
Title:			
Identification	1a	Identify the report as a protocol of a systematic review	1
Update	1b	If the protocol is for an update of a previous systematic review, identify as such	n.a.
Registration	2	If registered, provide the name of the registry (such as PROSPERO) and registration number	2
Authors:			
Contact	3a	Provide name, institutional affiliation, e-mail address of all protocol authors; provide physical mailing address of corresponding author	1
Contributions	3b	Describe contributions of protocol authors and identify the guarantor of the review	11
Amendments	4	If the protocol represents an amendment of a previously completed or published protocol, identify as such and list changes; otherwise, state plan for documenting important protocol amendments	n.a.
Support:			
Sources	5a	Indicate sources of financial or other support for the review	11
Sponsor	5b	Provide name for the review funder and/or sponsor	n.a.
Role of sponsor or funder	5c	Describe roles of funder(s), sponsor(s), and/or institution(s), if any, in developing the protocol	n.a.
INTRODUCTION			
Rationale	6	Describe the rationale for the review in the context of what is already known	2-4
Objectives	7	Provide an explicit statement of the **question(s) the review will address** with reference to participants, interventions, comparators, and outcomes (PICO)	6
METHODS			
Eligibility criteria	8	Specify the study characteristics (such as PICO, study design, setting, time frame) and report characteristics (such as years considered, language, publication status) to be used as criteria for eligibility for the review	6-7
Information sources	9	Describe all intended information sources (such as electronic databases, contact with study authors, trial registers or other grey literature sources) with planned dates of coverage	6-7
Search strategy	10	Present draft of search strategy to be used for at least one electronic database, including planned limits, such that it could be repeated	6, suppl. file 1

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Study records:			
Data management	11a	Describe the mechanism(s) that will be used to manage records and data throughout the review	7-9
Selection process	11b	State the process that will be used for selecting studies (such as two independent reviewers) through each phase of the review (that is, screening, eligibility and inclusion in meta-analysis)	7, Table 2
Data collection process	11c	Describe planned method of extracting data from reports (such as piloting forms, done independently, in duplicate), any processes for obtaining and confirming data from investigators	9
Data items	12	List and define all variables for which data will be sought (such as PICO items, funding sources), any pre-planned data assumptions and simplifications	6, table1
Outcomes and prioritization	13	List and define all outcomes for which data will be sought, including prioritization of main and additional outcomes, with rationale	5
Risk of bias in individual studies	14	Describe anticipated methods for assessing risk of bias of individual studies, including whether this will be done at the outcome or study level, or both; state how this information will be used in data synthesis	n.a.
Data synthesis	15a	Describe criteria under which study data will be quantitatively synthesised	n.a.
	15b	If data are appropriate for quantitative synthesis, describe planned summary measures, methods of handling data and methods of combining data from studies, including any planned exploration of consistency (such as I^2, Kendall's τ)	n.a.
	15c	Describe any proposed additional analyses (such as sensitivity or subgroup analyses, meta-regression)	n.a.
	15d	If quantitative synthesis is not appropriate, describe the type of summary planned	9
Meta-bias(es)	16	Specify any planned assessment of meta-bias(es) (such as publication bias across studies, selective reporting within studies)	n.a.
Confidence in cumulative evidence	17	Describe how the strength of the body of evidence will be assessed (such as GRADE)	n.a.

*** It is strongly recommended that this checklist be read in conjunction with the PRISMA-P Explanation and Elaboration (cite when available) for important clarification on the items. Amendments to a review protocol should be tracked and dated. The copyright for PRISMA-P (including checklist) is held by the PRISMA-P Group and is distributed under a Creative Commons Attribution Licence 4.0.**

From: Shamseer L, Moher D, Clarke M, Ghersi D, Liberati A, Petticrew M, Shekelle P, Stewart L, PRISMA-P Group. Preferred reporting items for systematic review and meta-analysis protocols (PRISMA-P) 2015: elaboration and explanation. BMJ. 2015 Jan 2;349(jan02 1):g7647.

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ENTREQ-Checklist Item 17-19

		page
Item 1-16		n.a.
17 Coding	Describe the process for coding of data (e.g. line by line coding to search for concepts)	4-5
18 Study comparison	Describe how were comparisons made within and across studies (e.g. subsequent studies were coded into pre-existing concepts, and new concepts were created when deemed necessary).	4-5
19 Derivation of themes	Explain whether the process of deriving the themes or constructs was inductive or deductive	4-5
Item 20-21		n.a.

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Healthcare related needs and desires of older people with post-stroke aphasia: study protocol of a systematic review and qualitative evidence synthesis using two different approaches

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Primary Subject Heading:	Geriatric medicine
Secondary Subject Heading:	Qualitative research
Keywords:	QUALITATIVE RESEARCH, Stroke < NEUROLOGY, STATISTICS & RESEARCH METHODS

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Healthcare related needs and desires of older people with post-stroke aphasia: study protocol of a systematic review and qualitative evidence synthesis using two different approaches

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Keywords: qualitative research, qualitative evidence synthesis, meta-ethnography, thematic synthesis, aphasia

Abstract

Introduction: Stroke is a frequent disease in the older population of Western Europe with aphasia as a common consequence. Aphasia is known to impede targeting treatment to individual partients' needs and therefore may reduce treatment success. In Germany, the post-acute care of stroke patients is provided by different health care institutions of different sectors (rehabilitation, nursing, primary care) with substantial difficulties to coordinate services. We will conduct two qualitative evidence syntheses aiming at exploring distinct health care needs and desires of older people living withpoststroke aphasia. We thereby hope to support the development of integrated care models based on needs of patients who are very restricted to communicate them. Since various methods of qualitative evidence syntheses exist, the aim of the study embedding the two qualitative evidence syntheses, is to determine if findings differ according to the approach used.

Methods and analysis: We will conduct two qualitative evidence syntheses by using meta-ethnography (ME) and thematic synthesis (ThS) independently to synthesize the findings of primary qualitative studies. The main differences between these two methods are the underlying epistemologies [idealism (ME) vs. realism (ThS)] and the type of research question [emerging (ME) vs. fixed (ThS)].

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We will search seven bibliographic databases. Inclusion criteria comprise: patients with post-stroke aphasia, aged 65 and older, studies in German/English, all types of qualitative studies concerning needs and desires related to health care or the health care system. The protocol was registered on PROSPERO, follows PRISMA-P guidelines and includes three items from the ENTREQ checklist.

Ethics and dissemination: Ethical approval is not required. Findings will be published in a peer-reviewed journal and presented on national conferences.

PROSPERO registration number: CRD42020171432.

Strengths and limitations of this study

Strengths:

- We will conduct the QES in an interdisciplinary research team with members highly experienced in qualitative research.
- All steps of the research process will be discussed both within our research team and with the interdisciplinary working group "Qualitative Methods in Health Services Research" of the German Network of Health Services Research.
- Study screening and data extraction will be conducted independently in two teams of two authors, and a fifth author mediating any disagreements.
- The conduction of two parallel syntheses will strengthen the validity of the findings and add to the methodological discussion on qualitative evidence synthesis.

Limitations:

- We will only consider publications in English and German language.

Introduction

Stroke is a frequent disease in Western European countries and the second most common cause of death in Germany and worldwide [1]. However, in Germany, epidemiological data on stroke rely on estimates derived from stroke registries or data from health insurance companies and therefore vary due to methodological differences [2]. Probably 243.000 to 260.000 persons per year are affected by stroke [2], with a clear tendency to risk increasing with age, especially for people aged 65 and older [3]. Aphasia is known to be a very common consequence of stroke: It occurs in about 30% of people with stroke [4]. There are no primary epidemiological data of prevalence and incidence of poststroke aphasia in Germany available. According to Huber et al. [5] in Germany each year about 25.000 people are newly affected by aphasia after stroke. Aphasic symptoms disappear frequently in the short term, but remain in approximately 56% of patients more than ten months after the stroke [6]. This has considerable impact on postacute treatment of stroke with interaction between patient and health care provider being affected by patients' communicative deficits. These deficits are even more important in a highly segregated care system with different logics and payment structures. While the merits of acute care for stroke are known, little is known about post-acute health care in stroke patients [7]. The coordination of care and the transition between sectors proves to be very challenging [8]. These transitions are prone to overburden patients, especially patients with aphasia, who literally have a problem to raise their voices. There is a

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substantial need for integrated care and to inform respective health care domains, especially rehabilitation, nursing and primary health care services.

Several quantitative investigations explored whether stroke survivors' needs are addressed adequately in the health care system [9–15]. All of them revealed a substantial proportion of individuals with unmet needs from at least 51% [13] up to 100 % [12, 16]. These studies clearly show that existing unmet needs are unfavourable related to the severity of disease or disability as well as to psychosocial aspects. For example Andrews et al. [10] showed that unfulfilled needs were associated with greater disability, fatigue and memory problems. Furthermore, data showed a relation to social status, as the risk for having unmet needs was associated with living in deprived areas, low income or being member of an ethnic minority group. Chen et al. [17] discovered, that on average 73.8% of stroke survivors have at least one unmet need. The types of needs mentioned in their systematic review ranged from needs related to body functions, activities and participation or environment to service needs in information, transport, home support / personal care and therapy.

We found three qualitative evidence syntheses (QES) related to our research question [18–20]: Hilari et al. [18] aimed at identifying prognostic factors of health-related quality of life of people with post-stroke aphasia, but not on their needs related to health care. Moreover, they included only three qualitative studies in their narrative synthesis. In their thematic analysis Manning et al. [19] did not focus on health care needs or health care experiences, but were interested more generally in factors contributing to personal recovery and living successfully with aphasia. The meta-ethnography by Pindus et al. [20] focussed on primary and community care and included all stroke patients independent of existing aphasia. Therefore, our review will contribute to answer the question of how health services in Germany could be better suited to the special needs and desires of older people with post-stroke aphasia.

Methods for synthesizing qualitative research: which one to choose when?

Our aim is not only to generate findings with respect to the needs of people with poststroke aphasia, but also to contribute to the methodological discussion on QES. Albeit theory-based recommendations for deciding which type of QES to choose for certain qualitative research questions exist [21–23], to date there is no *empirical* evidence, showing in which way and to what extent different methods of synthesizing lead to different challenges and discrepancies during the review process as well as in the findings. Of the QES methods presented in Booth et al. [21] we chose Meta-Ethnography (ME) [24] and Thematic Synthesis (ThS) [25] because of their ontological positions: ME for epistemological idealism and ThS for epistemological realism. In the spectrum between idealism and realism both methods are closely related [23]: ME as representative of objective idealism**,** that means seeing the world as a result of collectively shared understandings and ThS as a representative of critical realism which assumes, that our knowledge of the 'objective' reality is influenced by our perceptions and beliefs [23, 26]. Although from different points of view – one that assumes there is no reality 'per se' but peoples' shared experience creates reality, and one that assumes there is one reality, but

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its awareness is violated by our attitudes, abilities, and experiences – the common ground of both may be the idea of people negotiating what reality is.

With regard to the research question, we chose ME as an approach permitting adjustment of the research question in the course of the review conduction and ThS as an example for a method requiring a fixed research question [21].

Although different in terms of epistemology and research question, both methods are suited to go beyond the findings of primary studies. This is well known to be the case for ME [24, 27, 28], but also confirmed for ThS by Thomas & Harden [25] or Flemming et al. [29], who pointed out that ThS findings may as well generate new interpretations, constructs or hypotheses. The possibility to do so, mainly relies on the quality and depth of the primary studies [29].

Description of the two methods chosen for synthesizing the primary studies

1. Meta-Ethnography

Noblit & Hare developed meta-ethnography as a method to reach a higher level of analysis than one can expect from single studies and that may lead to new research questions [28]. They defined synthesis as an activity in which separate parts are brought together to form a 'whole' instead of a mere aggregation of data. This construction of the whole is characterized by some degree of innovation, so that the result is greater than the sum of its parts [24].

Meta-ethnography comprises seven phases: In the first phase, the interest of the research has to be determined. Afterwards it has to be decided, which research is relevant to the determined interest (phase 2) and the selected studies will be read (phase 3). This phase is followed by determining, how the studies are related (phase 4). According to Noblit and Hare, three different ways of relation are possible: 1. The studies are roughly similar, 2. the studies are contradictory, and 3. the studies build a line of argument [24]. According to their relation the studies are translated into one another (phase 5). The roughly similar studies will be referred to the reciprocal translation, which means that constructs, metaphors, categories etc. from similar studies are translated into each other. If the studies are contradictory, a refutational synthesis will be performed in which contradictory or conflicting findings are investigated, aiming to explain the differences between the single studies [23, 24, 28]. In phase 6 the translations are synthesized. This procedure should result in an overarching system of so-called third-order categories, that means: the interpretation of the reviewer [30]. Additionally, in this phase a line-of-argument (LOA) synthesis is done. This type of synthesis goes further than translation and permits a wider view on the topic. Although Noblit and Hare [24] describe the LOA-synthesis as one of three possible methods to be used in step 4, we agree with France et al. 2019 [27] that it should belong to the interpretive synthesis process. Phase 7 relies on reflections about where and in which form to publish the synthesis. Noblit & Hare point out that the publication should be prepared according to the intended audience and the findings achieved [24].

2. Thematic Synthesis

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Thematic synthesis is a method derived from thematic analysis and was developed by Thomas & Harden in the context of health promotion and public health research to address questions about intervention need, appropriateness and acceptability, and factors influencing intervention implementation. This approach shares some characteristics with meta-ethnography, e.g., in the reciprocal translation of descriptive and analytical themes. Moreover, there are similarities with grounded theory as the approach is inductive and themes are developed using a "constant comparison" method [25]. The synthesis process consists of three steps: "line-by-line"-coding (1), development of descriptive themes (2) and generating analytical themes (3) in the last step [25].

The first step is coding the whole text of the results/findings section of the primary study line-by-line and thereby translating concepts from one study to another. In step two new codes are created to capture the meaning of groups of initial codes (generation of descriptive themes). A synthesis of findings will be generated based on the findings of all original studies and organized by these descriptive themes. During discussion of findings' consistency, interrelations between codes and descriptive themes as well as a potential hierarchical order will be examined. The last step is the development of analytical themes, which is seen as an interpretative synthesis that goes beyond the content of the original studies. In this stage models or frameworks may be developed to identify and display relationships between the descriptive themes [25].

Objective

This protocol is part of a broader (comprehensive) research project adressing two questions. First: Which needs and desires in the context of post-acute care do people with post-stroke aphasia have? To answer this question, we undertake a systematic review of the existing qualitative literature using two different aproaches. This part is described here in detail. Beyond that, a second objective of our research is to figure out if, and to what extent, the processes and findings differ according to the respective aproach. The details of the procedures required to answer this question are going to be described in a forthcoming article.

Methods

The protocol was registered under CRD42020171432 with the International Prospective Register of Systematic Reviews (PROSPERO). Changes will be documented in PROSPERO, providing date and rationale for each modification.

The reporting of this protocol is in accordance with the *Preferred Reporting Items for Systematic Reviews and Meta-Analyses Statement for Protocols* (PRISMA-P [31]) and three additional items (item 17-19) from the *Enhancing Transparency in Reporting the synthesis of Qualitative research* (ENTREQ) checklist [32]). The full reports of the reviews will follow the PRISMA [33] and ENTREQ recommendations, the eMERGe guidance (for the meta-ethnography) [34] and partly the EPOC-criteria [35]. Moreover we will use the GRADE-CERQual-series to guide us through the procedure of assessing confidence in the evidence from the two reviews [36–40].

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Patient and Public Involvement

No patient involved.

Search for similar review projects

To verify that our QES are embedded in current research we searched for published and unpublished QES concerning our topic. We searched Medline via PubMed for published and PROSPERO for unpublished QES and found three reviews [18–20] (see introduction section for further details).

PICoS

We chose the PICo tool [41] to formulate our research question. PICo in qualitative research means: **P**opulation, Phenomenon of **I**nterest, **Co**ntext. Our objective is to identify studies that explore health care needs and desires (phenomenon of interest) of people with poststroke aphasia (population) during postacute treatment (context). We used the PICo-scheme complemented by requirements for the **s**tudy design to guide search term selection and inclusion criteria (so-called PICoS-scheme, see also table 1).

Search

Databases and search terms

Search strategies aimed at identifying qualitative research were developed by an experienced Cochrane Information Specialist using an approach guided by relevant literature. First, a preliminary search strategy guided by the PICoS-scheme using text words and controlled vocabulary (Medical Subject Headings) was designed for MEDLINE. Second, twelve relevant articles meeting the inclusion criteria were identified by conducting and screening a scoping search (see Supplementary File 1). These articles were subsequently used to test and refine the MEDLINE strategy. The test revealed that the concept 'stroke' is not consistently mentioned in the abstracts of relevant publications. Therefore, the concept was omitted from the search strategy to achieve the highest possible sensitivity. The resulting search strategy consists of a search block for 'aphasia' and is combined with validated and sensitive search filters to identify qualitative research, where available for the respective database. The refined MEDLINE strategy was adapted to PsycINFO, Psyindex, CINAHL, Web of Science (Social Science Citation Index), ASSIA and a database indexing grey literature: BASE. Supplementary File 1 provides details of all search strategies.

Additional search strategies

In addition to the database search, the review authors will screen the reference lists of included studies.

Inclusion and exclusion criteria

We will use the criteria outlined in table 1 to select appropriate studies. As health care needs differ according to age [42] and most people affected by aphasia after stroke are of advanced age [43], our study focuses on people aged 65 and older.

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Table 1: Inclusion and exclusion criteria for study selection

	Inclusion criteria	Exclusion criteria
Population	Adult People, aged 65 and older, with post-stroke aphasia	Study includes only members of particular (minority) groups
Interest of the study	Needs and desires related to (the) health care (system)	Study focusses special needs related to technical devices Studies examining certain therapeutic means
Context of the study	In- or outpatient setting, post-acute treatment	
Study design	All types of primary qualitative studies (= no reviews), mixed method studies only if sufficient qualitative data provided (e.g. separate qualitative data analysis)	Single-case studies
Languages	German, English	
Publication status	Published or unpublished, e-first, full text available	
Type of publication	Journal article or book chapter	

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Selection of relevant studies

A PRISMA compliant flow chart [33] will be used to document and summarize the study selection process. Search results will be uploaded to RAYYAN [44], a free software tool allowing reviewer teams to collaborate on screening references online. The selection of relevant studies will be performed in groups of at least two reviewers and will comprise the steps listed in table 2: Parallel to the identification of content-related in- and exclusion criteria, method-specific in- and exclusion criteria will be gathered from the methodological literature (step 1). Content-related criteria are used to screen the search results and establish a list of preliminary included studies (step 2). The full-text of the studies will be read in order to remove studies not fulfilling the content-related inclusion criteria in detail (step 3). The resulting sample from this step will be examined using the method-specific criteria (step 4). Neither of the review authors will be blind to the journal titles or to the study authors or the institutions.

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Table 2: Selection process (Modified according to [33])

1	Identification	**Objective:**	Building a consensus-based framework of content-related inclusion and exclusion criteria and gathering method-specific criteria
		Method:	Application and refining of content-related in- and exclusion criteria, formulation of method-specific in- and exclusion criteria
		Material:	Titles and abstracts of a random sample of 100 articles for content-related and methodological literature for method-specific criteria
		▪ A subset of 100 articles randomly selected out of the search hits, will be uploaded to RAYYAN ▪ Each reviewer screens all of the 100 articles. ▪ All reviewers compare their results. ▪ Disagreeing judgements will be discussed and inclusion criteria will be refined through this process. ▪ Pairs of researchers (VLev&YE, VLen&NP) derive method-specific in- and exclusion criteria from the methodological literature. ▪ The consensus process and the modification of content-related and method-specific inclusion / exclusion criteria (if done) will be documented.	
2	Screening	**Objective:**	Consensus-based list of preliminary included studies based on content-related criteria
		Method:	Screening of all records using content-related in- and exclusion criteria
		Material:	Abstracts of all articles retrieved
		▪ All hits are screened and evaluated on the basis of the finally consented content-related criteria. ▪ The procedure will be done in two pairs of two reviewers (VLev&YE, VLen&NP). Each pair evaluates half of the hits. ▪ The respective pair of reviewers will meet to discuss and compare their lists of preliminary included studies. ▪ If no consensus is reached, a third reviewer (TM) will read the abstract and moderate the consensus process. ▪ The two lists of preliminary studies (one list per pair) will be joined to one. ▪ If modifications of content-related inclusion / exclusion criteria are necessary, this will be done and be documented.	
3	Eligibility	**Objective:**	Consensus-based list of finally included studies based on content-related criteria, list of excluded studies (incl. reason for exclusion)
		Method:	Full-text screening of the preliminary included studies using content-related in- and exclusion criteria
		Material:	Full text
		▪ All reviewers will read the full text of the preliminary included studies. ▪ Each reviewer creates a list of finally included and a list of excluded studies (with documentation of reason for exclusion). ▪ Consensus will be sought regarding in- and exclusion of studies. The process and decisions will be documented. ▪ The consensus process will result in one list of finally included and one list of excluded studies (with documentation of reason for exclusion).	
4	Inclusion	**Objective:**	Method-specific**, c**onsensus-based lists of finally included studies, lists of excluded studies for each method (incl. reason for exclusion)
		Method:	Full-text screening of the studies included in step 3 using method-specific in- and exclusion criteria
		Material:	Full text
		▪ Method-specific review teams will read the full text of the studies included in step 3 ▪ Each reviewer creates a list of studies to be included on the basis of method-specific criteria (with documentation of reason for exclusion). ▪ Consensus about in- and exclusion will be sought in the method-specific teams. The process and decisions will be documented. ▪ This process will result in two lists of studies to be in- or excluded	

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Critical appraisal of study quality

Noblit & Hare [24] did not take a stance on quality appraisal of the primary studies included in meta-ethnography. Other authors of meta-ethnographic methodological research present contrasting opinions on quality appraisal: Whereas Atkins et al. [28], Campbell et al. [45], Froud et al. [46], Pound et al. [47] or Toye et al. [48] seem to be willing to explore the methodological quality of studies to be included in meta-ethnographies, Britten et al. [30] or France et al. [27] point out, that time could be spent more efficiently by concentrating on the topic of the review [30] or that critical appraisal is redundant as long as it has no implications for in- or exclusion of studies [27]. Thomas & Harden performed critical appraisal of the primary studies included in their thematic synthesis and undertook a sensitivity analysis to determine the impact of low-quality studies on the findings of the review [25]. Overall, the position of qualitative researchers on this topic is still indecisive. What convinced us to include critical appraisal, were the findings of Pindus et al. that a good quality score does not necessarily mean, that the study provides sufficient contextual data [20], and France et al. [27], that poor descriptions of methods do not necessarily equate to a poorly-conducted study. Toye et al. [49] drew attention to the fact, that insightfulness and methodological rigour are different facets of quality and may be quite different. So, we felt that the determination of methodological quality could contribute to the ongoing debate, as long as we go beyond the mere exclusion of methodologically flawed studies [49, 50]. We will follow Thomas & Harden [25] and their sensitivity analysis approach and further will explore the relation between insightfulness and methodological quality in the included studies. To determine the methodological quality of the included studies we chose the widely used Critical Appraisal Skills Programme (CASP) Checklist for qualitative studies [37, 51, 52].

Data extraction and analysis

For meta-ethnography as well as for thematic synthesis researcher pairs will independently extract data into spreadsheets and resolve potential discrepancies by discussion. Unsolved disagreements will be discussed with a fifth researcher. From studies employing mixed methods designs only qualitative data will be extracted. Extracted background data for each study include: study-design, objective, location, setting, participants and method of data collection and analysis.

For the meta-ethnographic analysis second order constructs on subjective needs and desires related to health care will be extracted. Second order constructs are defined as the interpretation of the authors of the primary study [48]. First order constructs (the interpretations of the participants of the primary studies) are to be additionally included in our analysis only if clarification of the second order constructs is required [27]. The second order constructs will lead to third order constructs, in other words: the interpretation of the reviewer team [25, 30]. For the thematic synthesis all of the text in the results/findigs section is to be analyzed in three steps: First, a line-by-line coding, secondly the organization of the resulting codes to descriptive themes and thirdly the development of analytical themes out of the descriptive ones. These analytical themes

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correspond to the third order constructs of the meta-ethnographical analysis [25] and therefore can be used to compare the findings of ME and ThS.

Constant comparison process

Two independent working groups will each conduct a review using one of the outlined methods. Beyond that, our intention is to compare the challenges, questions, and findings arising from each step of the review process to generate empirical material that may clarify the similarities as well as the differences of the two methods. For there is no existing framework that could guide us through the process, we will develop the distinct steps of the comparison during the review process using the method of constant comparison. As an example, we describe how we will proceed if the selection process ends up in two different study samples: To avoid that further differences, particularly the findings of the synthesis in terms of third order constructs / analytical themes, are seen as a result of the divergent study samples, we will analyse both samples with each method. That means: First each group analyses the studies fulfilling the content-related *and* the specific inclusion criteria of 'their' method, and afterwards they include the method-specific studies of the other group – an approach very similar to sensitivity analysis suggested by Thomas & Harden [25]. This will enable us to distinguish between differences resulting from the methodological steps applied and differences based only on the fact, that different studies were included in the QES.

The findings of the two approaches are to be compared with respect to the process of conduction; the resulting third order constructs (ME) / analytical themes (ThS), the scope of the findings and their potential of generalization. Because we expect challenges to occur while performing the particular steps of the review, we decided to determine the exact procedures during the process – a strategy named "Gegenstandsangemessenheit" and seen as an indicator of quality in qualitative research [53–55]. To make this process visible, we will document the respective problem, our discussion and the way we decide to solve it – including the reason for the decision - in a forthcoming methodological article.

A systematic comparison between two qualitative synthesis approaches cannot limit itself to be descriptive, but will have to be interpretive to explain the differences. The point of departure of this analysis will be a comparison of the findings of both approaches. Differences and similarities will be identified. In a second step we will try to understand the differences by looking closer at the respective methods, processes and decisions made during the conduct of the respective study. This will enable us to identify reasons for decisions made and to distinguish between possible reasons that are grounded in the respective methodology or approach vs. other reasons, such as personal backgrounds of the reviewers or special theoretical stances. By doing this we make reference to the comprehensive meta-study approach and plan to explain differences between the findings of the two approaches due to reasons related to data analysis, methods and theories applied [56].

Author background and reflexivity

All participating reviewers are experienced qualitative researchers, with different clinical and scientific backgrounds: psychology / health services research, lecturer for qualitative

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research methods, papers on e.g. qualitative data acquisition and analysis (NP); psychology / rehabilitation sciences and health services research; lecturer in interpretative qualitative methods, papers on e.g. quality criteria of qualitative research and qualitative sampling (TM); gerontology / palliative care (VLev, YE) and Public Health / nursing / physical therapy, lecturer for qualitative and health services research (VLen). This will enable us to adopt a comprehensive view on the topic investigated. Constant discussion and regular meetings will help to avoid skewing the analysis and interpretation and thus minimize reviewer bias. Potential threats to credibility as well as reviewer bias will also be minimized by rigorous and transparent discussion in our interdisciplinary research group "Qualitative Methods in Health Services Research" of the German Network of Health-Services Research, as well as determination and documentation of all procedures during the review process. To disclose how our positions may have influenced the review and whether our attitudes toward the research topic changed during the review, we will include a reflexive statement [35] in the results section of our review.

Discussion

By using two rigorous and codified methods (ME, ThS) for synthesizing qualitative research, we expect to generate in-depth conceptual understanding of what is important for people with poststroke aphasia. The identification of all variations of patients' (living with aphasia) needs and desires related to health care may contribute to tailor health care in a more patient-centred way – which is known to improve therapeutic outcomes [57–59]. The combination of two different QES methods is unique to our study. As far as we know such type of review has never been conducted before and it will allow insights on the impact the QES methods may have not only on the findings but on the whole review process.

A limitation of our study is that we will not include publications in languages other than German or English. This might limit the range of the findings of our QES.

In addition, a comparison of different QES methods can only be made with some methodological limitations. From a pure conceptual or even idealistic perspective, a strict comparison is hardly possible or only possible for two quite similar approaches. We chose two approaches similar enough to answer the same research question (leaving it open to adapt the question during the course of the study if necessary). We assume, however, that the reality of choice over QES approaches is in practice much more dependent on criteria that go beyond methodological considerations – such as personal background and familiarity with a method, personal preferences of team members etc. [21]. In practice, therefore, boundaries between different QES approaches might be blurred with regard to concepts, methods or terms. Since there is no *ideal* comparison in this vein, we find it appropriate to make a few restrictions to the respective QES methodologies in order to allow for substantive comparisons. It is well acknowledged that these restrictions should be an explicit part of the analysis, i.e. the approach to comparison.

Ethics and dissemination

Ethical approval is not required, because no human subjects or personal data are directly involved in our review. To ensure dissemination, the findings of the QES will be published

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in an international peer-reviewed journal and presented on national conferences. A distinct method paper will be published in an international peer-reviewed journal.

Authors' contributions: NP is the guarantor. VLen and NP drafted the manuscript. MIM designed the search strategy. All authors (NP, TM, YE, MIM, VLev, VLen) read, provided feedback on and approved the final manuscript.

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Supplementary File 1: Search strategies and relevant references used to test the MEDLINE strategy

Medline (PubMed)

(comprises all terms of "HIRU hedges Qualitative – MEDLINE filter, best balance of sensitivity and specificity" and MeSH and textwords identified during text analysis of twelve relevant references, adding sensitivity)

((Aphasia[mh] OR aphasia*[tw]) AND (Qualitative Research[mh] OR Interviews as Topic[mh] OR Interview[pt] OR Focus Groups[mh] OR experience*[tw] OR qualitative[tw] OR interview*[tw] OR needs[tw] OR themes[tw] OR perspective*[tw]))

PsycINFO (Ovid SP)

1. Aphasia/
2. aphasia*.tw.
3. 1 or 2

[HIRU hedges Qualitative – PsycINFO filter, best balance of sensitivity and specificity]

4. experiences.tw. or interview*.tw. or qualitative*.tw.
5. 3 and 4

Psyndex (PubPsych)

(aphasi* AND (experience* OR qualitativ* OR interview* OR needs OR themes OR perspective*)) AND DB="PSYNDEX"

CINAHL (EBSCO)

1. MH "aphasia+"
2. TI (aphasia*) OR AB (aphasia*)
3. S1 OR S2

[Wilczynski 2007 "small drop sensitivity"-filter]

4. MH "attitude+" OR MH "interviews+" OR MH "qualitative studies"
5. S3 AND S4
6. MH "Interview+" OR MH "Grounded theory" OR MH "Focus Groups" OR MH "Field Studies" OR MH "Life Experiences+"
7. S3 AND S6
8. S5 OR S7

Wilczynski NL, Marks S, Haynes RB. Search strategies for identifying qualitative studies in CINAHL. Qual Health Res. 2007 May;17(5):705-10.

Social Science Citation Index (Web of Science)

((TS=aphasia*) AND (TS=qualitative OR TS=interview*))

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ASSIA (ProQuest)

ALL(aphasia*) AND ALL(qualitative OR interview* OR focus group* OR experience* OR need* OR theme* OR perspective* OR attitude*)

Bielefeld Academic Search Engine (BASE – Advanced Search)

[limited to doctype = Dissertationen und postgraduale Arbeiten]

(aphasia*) AND (qualitative OR interview* OR experience* OR needs OR themes OR perspective*) doctype:183

Twelve relevant references used to test the MEDLINE search strategy

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ENTREQ-Checklist Item 17-19

		page
Item 1-16		n.a.
17 Coding	Describe the process for coding of data (e.g. line by line coding to search for concepts)	4-5
18 Study comparison	Describe how were comparisons made within and across studies (e.g. subsequent studies were coded into pre-existing concepts, and new concepts were created when deemed necessary).	4-5
19 Derivation of themes	Explain whether the process of deriving the themes or constructs was inductive or deductive	4-5
Item 20-21		n.a.

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PRISMA-P (Preferred Reporting Items for Systematic review and Meta-Analysis Protocols) 2015 checklist: recommended items to address in a systematic review protocol*

Section and topic	Item No	Checklist item	page
ADMINISTRATIVE INFORMATION			
Title:			
Identification	1a	Identify the report as a protocol of a systematic review	1
Update	1b	If the protocol is for an update of a previous systematic review, identify as such	n.a.
Registration	2	If registered, provide the name of the registry (such as PROSPERO) and registration number	2
Authors:			
Contact	3a	Provide name, institutional affiliation, e-mail address of all protocol authors; provide physical mailing address of corresponding author	1
Contributions	3b	Describe contributions of protocol authors and identify the guarantor of the review	11
Amendments	4	If the protocol represents an amendment of a previously completed or published protocol, identify as such and list changes; otherwise, state plan for documenting important protocol amendments	n.a.
Support:			
Sources	5a	Indicate sources of financial or other support for the review	11
Sponsor	5b	Provide name for the review funder and/or sponsor	n.a.
Role of sponsor or funder	5c	Describe roles of funder(s), sponsor(s), and/or institution(s), if any, in developing the protocol	n.a.
INTRODUCTION			
Rationale	6	Describe the rationale for the review in the context of what is already known	2-4
Objectives	7	Provide an explicit statement of the **question(s) the review will address** with reference to participants, interventions, comparators, and outcomes (PICO)	6
METHODS			
Eligibility criteria	8	Specify the study characteristics (such as PICO, study design, setting, time frame) and report characteristics (such as years considered, language, publication status) to be used as criteria for eligibility for the review	6-7
Information sources	9	Describe all intended information sources (such as electronic databases, contact with study authors, trial registers or other grey literature sources) with planned dates of coverage	6-7
Search strategy	10	Present draft of search strategy to be used for at least one electronic database, including planned limits, such that it could be repeated	6, suppl. file 1

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Study records:			
Data management	11a	Describe the mechanism(s) that will be used to manage records and data throughout the review	7-9
Selection process	11b	State the process that will be used for selecting studies (such as two independent reviewers) through each phase of the review (that is, screening, eligibility and inclusion in meta-analysis)	7, Table 2
Data collection process	11c	Describe planned method of extracting data from reports (such as piloting forms, done independently, in duplicate), any processes for obtaining and confirming data from investigators	9
Data items	12	List and define all variables for which data will be sought (such as PICO items, funding sources), any pre-planned data assumptions and simplifications	6, table1
Outcomes and prioritization	13	List and define all outcomes for which data will be sought, including prioritization of main and additional outcomes, with rationale	5
Risk of bias in individual studies	14	Describe anticipated methods for assessing risk of bias of individual studies, including whether this will be done at the outcome or study level, or both; state how this information will be used in data synthesis	n.a.
Data synthesis	15a	Describe criteria under which study data will be quantitatively synthesised	n.a.
	15b	If data are appropriate for quantitative synthesis, describe planned summary measures, methods of handling data and methods of combining data from studies, including any planned exploration of consistency (such as I^2, Kendall's τ)	n.a.
	15c	Describe any proposed additional analyses (such as sensitivity or subgroup analyses, meta-regression)	n.a.
	15d	If quantitative synthesis is not appropriate, describe the type of summary planned	9
Meta-bias(es)	16	Specify any planned assessment of meta-bias(es) (such as publication bias across studies, selective reporting within studies)	n.a.
Confidence in cumulative evidence	17	Describe how the strength of the body of evidence will be assessed (such as GRADE)	n.a.

*** It is strongly recommended that this checklist be read in conjunction with the PRISMA-P Explanation and Elaboration (cite when available) for important clarification on the items. Amendments to a review protocol should be tracked and dated. The copyright for PRISMA-P (including checklist) is held by the PRISMA-P Group and is distributed under a Creative Commons Attribution Licence 4.0.**

From: Shamseer L, Moher D, Clarke M, Ghersi D, Liberati A, Petticrew M, Shekelle P, Stewart L, PRISMA-P Group. Preferred reporting items for systematic review and meta-analysis protocols (PRISMA-P) 2015: elaboration and explanation. BMJ. 2015 Jan 2;349(jan02 1):g7647.

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Study protocol of a systematic review and qualitative evidence synthesis using two different approaches: Healthcare related needs and desires of older people with post-stroke aphasia

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Keywords:	QUALITATIVE RESEARCH, Stroke < NEUROLOGY, STATISTICS & RESEARCH METHODS

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Study protocol of a systematic review and qualitative evidence synthesis using two different approaches: Healthcare related needs and desires of older people with post-stroke aphasia

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Keywords: qualitative research, qualitative evidence synthesis, meta-ethnography, thematic synthesis, aphasia

Abstract

Introduction: Stroke is a frequent disease in the older population of Western Europe with aphasia as a common consequence. Aphasia is known to impede targeting treatment to individual partients' needs and therefore may reduce treatment success. In Germany, the post-acute care of stroke patients is provided by different health care institutions of different sectors (rehabilitation, nursing, primary care) with substantial difficulties to coordinate services. We will conduct two qualitative evidence syntheses aiming at exploring distinct health care needs and desires of older people living withpoststroke aphasia. We thereby hope to support the development of integrated care models based on needs of patients who are very restricted to communicate them. Since various methods of qualitative evidence syntheses exist, the aim of the study embedding the two qualitative evidence syntheses, is to determine if findings differ according to the approach used.

Methods and analysis: We will conduct two qualitative evidence syntheses by using meta-ethnography (ME) and thematic synthesis (ThS) independently to synthesize the findings of primary qualitative studies. The main differences between these two methods are the underlying epistemologies [idealism (ME) vs. realism (ThS)] and the type of research question [emerging (ME) vs. fixed (ThS)].

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We will search seven bibliographic databases. Inclusion criteria comprise: patients with post-stroke aphasia, aged 65 and older, studies in German/English, all types of qualitative studies concerning needs and desires related to health care or the health care system. The protocol was registered on PROSPERO, follows PRISMA-P guidelines and includes three items from the ENTREQ checklist.

Ethics and dissemination: Ethical approval is not required. Findings will be published in a peer-reviewed journal and presented on national conferences.

PROSPERO registration number: CRD42020171432.

Strengths and limitations of this study

- We will conduct the QES in an interdisciplinary research team with members highly experienced in qualitative research.
- All steps of the research process will be discussed both within our research team and with the interdisciplinary working group "Qualitative Methods in Health Services Research" of the German Network of Health Services Research.
- Study screening and data extraction will be conducted independently in two teams of two authors, and a fifth author mediating any disagreements.
- The conduction of two parallel syntheses will strengthen the validity of the findings and add to the methodological discussion on qualitative evidence synthesis.
- We will only consider publications in English and German language.

Introduction

Stroke is a frequent disease in Western European countries and the second most common cause of death in Germany and worldwide [1]. However, in Germany, epidemiological data on stroke rely on estimates derived from stroke registries or data from health insurance companies and therefore vary due to methodological differences [2]. Probably 243.000 to 260.000 persons per year are affected by stroke [2], with a clear tendency to risk increasing with age, especially for people aged 65 and older [3].

Aphasia is known to be a very common consequence of stroke: It occurs in about 30% of people with stroke [4]. There are no primary epidemiological data of prevalence and incidence of poststroke aphasia in Germany available. According to Huber et al. [5] in Germany each year about 25.000 people are newly affected by aphasia after stroke. Aphasic symptoms disappear frequently in the short term, but remain in approx. 56% of patients more than ten months after stroke [6]. This has considerable impact on postacute treatment of stroke with interaction between patient and health care provider being affected by patients' communicative deficits. These deficits are even more important in a highly segregated care system with different logics and payment structures.

While the merits of acute care for stroke are known, little is known about post-acute health care in stroke patients [7]. The coordination of care and the transition between sectors proves to be very challenging [8]. These transitions are prone to overburden patients, especially patients with aphasia, who literally have a problem to raise their voices. There is a substantial need for integrated care and to inform respective health care domains, especially rehabilitation, nursing and primary health care services.

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Quantitative studies of stroke survivors' needs [9–16] have revealed a substantial proportion of unmet health care needs ranging from at least 51% [14] up to 100 % [12, 13]. These studies clearly show that existing unmet needs are unfavourable related to the severity of disease or disability as well as to psychosocial aspects. For example Andrews et al. [10] showed that unfulfilled needs were associated with greater disability, fatigue and memory problems. Furthermore, data showed a relation to social status, as the risk for having unmet needs was associated with living in deprived areas, low income or being member of an ethnic minority group. Chen et al. [17] discovered, that on average 73.8% of stroke survivors have at least one unmet need. The types of needs mentioned in their systematic review ranged from needs related to body functions, activities and participation or environment to service needs in information, transport, home support / personal care and therapy.

We found three qualitative evidence syntheses (QES) related to our research question [18–20]: Hilari et al. [18] aimed at identifying prognostic factors of health-related quality of life of people with post-stroke aphasia, but not on their needs related to health care. Moreover, they included only three qualitative studies in their narrative synthesis. In their thematic analysis Manning et al. [19] did not focus on health care needs or health care experiences, but were interested more generally in factors contributing to personal recovery and living successfully with aphasia. The meta-ethnography by Pindus et al. [20] focussed on primary and community care and included all stroke patients independent of existing aphasia. Therefore, our review will contribute to answer the question of how health services in Germany could be better suited to the special needs and desires of older people with post-stroke aphasia.

Methods for synthesizing qualitative research: which one to choose when?

Our aim is not only to generate findings with respect to the needs of people with poststroke aphasia, but also to contribute to the methodological discussion on QES. Albeit theory-based recommendations for deciding which type of QES to choose for certain qualitative research questions exist [21–23], to date there is no *empirical* evidence, showing in which way and to what extent different methods of synthesizing lead to different challenges and discrepancies during the review process as well as in the findings. Of the QES methods presented in Booth et al. [21] we chose Meta-Ethnography (ME) [24] and Thematic Synthesis (ThS) [25] because of their ontological positions: ME for epistemological idealism and ThS for epistemological realism. In the spectrum between idealism and realism both methods are closely related [23]: ME as representative of objective idealism**,** that means seeing the world as a result of collectively shared understandings and ThS as a representative of critical realism which assumes, that our knowledge of the 'objective' reality is influenced by our perceptions and beliefs [23, 26]. Although from different points of view – one that assumes there is no reality 'per se' but peoples' shared experience creates reality, and one that assumes there is one reality, but its awareness is violated by our attitudes, abilities, and experiences – the common ground of both may be the idea of people negotiating what reality is.

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With regard to the research question, we chose ME as an approach permitting adjustment of the research question in the course of the review conduction and ThS as an example for a method requiring a fixed research question [21].

Although different in terms of epistemology and research question, both methods are suited to go beyond the findings of primary studies. This is well known to be the case for ME [24, 27, 28], but also confirmed for ThS by Thomas & Harden [25] or Flemming et al. [29], who pointed out that ThS findings may as well generate new interpretations, constructs or hypotheses. The possibility to do so, mainly relies on the quality and depth of the primary studies [29].

Description of the two methods chosen for synthesizing the primary studies

1. Meta-Ethnography

Noblit & Hare developed meta-ethnography as a method to reach a higher level of analysis than one can expect from single studies and that may lead to new research questions [27]. They defined synthesis as an activity in which separate parts are brought together to form a 'whole' instead of a mere aggregation of data. This construction of the whole is characterized by some degree of innovation, so that the result is greater than the sum of its parts [24].

Meta-ethnography comprises seven phases: In the first phase, the interest of the research has to be determined. Afterwards it has to be decided, which research is relevant to the determined interest (phase 2) and the selected studies will be read (phase 3). This phase is followed by determining, how the studies are related (phase 4). According to Noblit and Hare, three different ways of relation are possible: 1. The studies are roughly similar, 2. the studies are contradictory, and 3. the studies build a line of argument [24]. According to their relation the studies are translated into one another (phase 5). The roughly similar studies will be referred to the reciprocal translation, which means that constructs, metaphors, categories etc. from similar studies are translated into each other. If the studies are contradictory, a refutational synthesis will be performed in which contradictory or conflicting findings are investigated, aiming to explain the differences between the single studies [23, 24, 27]. In phase 6 the translations are synthesized. This procedure should result in an overarching system of so-called third-order categories, that means: the interpretation of the reviewer [30]. Additionally, in this phase a line-of-argument (LOA) synthesis is done. This type of synthesis goes further than translation and permits a wider view on the topic. Although Noblit and Hare [24] describe the LOA-synthesis as one of three possible methods to be used in step 4, we agree with France et al. 2019 [28] that it should belong to the interpretive synthesis process. Phase 7 relies on reflections about where and in which form to publish the synthesis. Noblit & Hare point out that the publication should be prepared according to the intended audience and the findings achieved [24].

2. Thematic Synthesis

Thematic synthesis is a method derived from thematic analysis and was developed by Thomas & Harden in the context of health promotion and public health research to

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address questions about intervention need, appropriateness and acceptability, and factors influencing intervention implementation. This approach shares some characteristics with meta-ethnography, e.g., in the reciprocal translation of descriptive and analytical themes. Moreover, there are similarities with grounded theory as the approach is inductive and themes are developed using a "constant comparison" method [25]. The synthesis process consists of three steps: "line-by-line"-coding (1), development of descriptive themes (2) and generating analytical themes (3) in the last step [25].

The first step is coding the whole text of the results/findings section of the primary study line-by-line and thereby translating concepts from one study to another. In step two new codes are created to capture the meaning of groups of initial codes (generation of descriptive themes). A synthesis of findings will be generated based on the findings of all original studies and organized by these descriptive themes. During discussion of findings' consistency, interrelations between codes and descriptive themes as well as a potential hierarchical order will be examined. The last step is the development of analytical themes, which is seen as an interpretative synthesis that goes beyond the content of the original studies. In this stage models or frameworks may be developed to identify and display relationships between the descriptive themes [25].

Objective

This protocol is part of a broader (comprehensive) research project adressing two questions. First: Which needs and desires in the context of post-acute care do people with post-stroke aphasia have? To answer this question, we undertake a systematic review of the existing qualitative literature using two different aproaches. This part is described here in detail. Beyond that, a second objective of our research is to figure out if, and to what extent, the processes and findings differ according to the respective aproach. The details of the procedures required to answer this question are going to be described in a forthcoming article.

Methods

The protocol was registered under CRD42020171432 with the International Prospective Register of Systematic Reviews (PROSPERO). Changes will be documented in PROSPERO, providing date and rationale for each modification.

The reporting of this protocol is in accordance with the *Preferred Reporting Items for Systematic Reviews and Meta-Analyses Statement for Protocols* (PRISMA-P [31]) and three additional items (item 17-19) from the *Enhancing Transparency in Reporting the synthesis of Qualitative research* (ENTREQ) checklist [32]). The full reports of the reviews will follow the PRISMA [33] and ENTREQ recommendations, the eMERGe guidance (for the meta-ethnography) [34] and partly the EPOC-criteria [35]. Moreover we will use the GRADE-CERQual-series to guide us through the procedure of assessing confidence in the evidence from the two reviews [36–40].

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Patient and Public Involvement

No patient involved.

Search for similar review projects

To verify that our QES are embedded in current research we searched for published and unpublished QES concerning our topic. We searched Medline via PubMed for published and PROSPERO for unpublished QES and found three reviews [18–20] (see introduction section for further details).

PICoS

We chose the PICo tool [41] to formulate our research question. PICo in qualitative research means: **P**opulation, Phenomenon of **I**nterest, **Co**ntext. Our objective is to identify studies that explore health care needs and desires (phenomenon of interest) of people with poststroke aphasia (population) during postacute treatment (context). We used the PICo-scheme complemented by requirements for the **s**tudy design to guide search term selection and inclusion criteria (so-called PICoS-scheme, see also table 1).

Search

Databases and search terms

Search strategies aimed at identifying qualitative research were developed by an experienced Cochrane Information Specialist using an approach guided by relevant literature. First, a preliminary search strategy guided by the PICoS-scheme using text words and controlled vocabulary (Medical Subject Headings) was designed for MEDLINE. Second, twelve relevant articles meeting the inclusion criteria were identified by conducting and screening a scoping search (see Supplementary File 1). These articles were subsequently used to test and refine the MEDLINE strategy. The test revealed that the concept 'stroke' is not consistently mentioned in the abstracts of relevant publications. Therefore, the concept was omitted from the search strategy to achieve the highest possible sensitivity. The resulting search strategy consists of a search block for 'aphasia' and is combined with validated and sensitive search filters to identify qualitative research, where available for the respective database. The refined MEDLINE strategy was adapted to PsycINFO, Psyindex, CINAHL, Web of Science (Social Science Citation Index), ASSIA and a database indexing grey literature: BASE. Supplementary File 1 provides details of all search strategies.

Additional search strategies

In addition to the database search, the review authors will screen the reference lists of included studies.

Inclusion and exclusion criteria

We will use the criteria outlined in table 1 to select appropriate studies. As health care needs differ according to age [42] and most people affected by aphasia after stroke are of advanced age [43], our study focuses on people aged 65 and older.

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Table 1: Inclusion and exclusion criteria for study selection

	Inclusion criteria	Exclusion criteria
Population	Adult People, aged 65 and older, with post-stroke aphasia	Study includes only members of particular (minority) groups
Interest of the study	Needs and desires related to (the) health care (system)	Study focusses special needs related to technical devices Studies examining certain therapeutic means
Context of the study	In- or outpatient setting, post-acute treatment	
Study design	All types of primary qualitative studies (= no reviews), mixed method studies only if sufficient qualitative data provided (e.g. separate qualitative data analysis)	Single-case studies
Languages	German, English	
Publication status	Published or unpublished, e-first, full text available	
Type of publication	Journal article or book chapter	

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Selection of relevant studies

A PRISMA compliant flow chart [33] will be used to document and summarize the study selection process. Search results will be uploaded to RAYYAN [44], a free software tool allowing reviewer teams to collaborate on screening references online. The selection of relevant studies will be performed in groups of at least two reviewers and will comprise the steps listed in table 2: Parallel to the identification of content-related in- and exclusion criteria, method-specific in- and exclusion criteria will be gathered from the methodological literature (step 1). Content-related criteria are used to screen the search results and establish a list of preliminary included studies (step 2). The full-text of the studies will be read in order to remove studies not fulfilling the content-related inclusion criteria in detail (step 3). The resulting sample from this step will be examined using the method-specific criteria (step 4). Neither of the review authors will be blind to the journal titles or to the study authors or the institutions.

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Table 2: Selection process (Modified according to [33])

1	Identification	**Objective:**	Building a consensus-based framework of content-related inclusion and exclusion criteria and gathering method-specific criteria
		Method:	Application and refining of content-related in- and exclusion criteria, formulation of method-specific in- and exclusion criteria
		Material:	Titles and abstracts of a random sample of 100 articles for content-related and methodological literature for method-specific criteria
		▪ A subset of 100 articles randomly selected out of the search hits, will be uploaded to RAYYAN ▪ Each reviewer screens all of the 100 articles. ▪ All reviewers compare their results. ▪ Disagreeing judgements will be discussed and inclusion criteria will be refined through this process. ▪ Pairs of researchers (VLev&YE, VLen&NP) derive method-specific in- and exclusion criteria from the methodological literature. ▪ The consensus process and the modification of content-related and method-specific inclusion / exclusion criteria (if done) will be documented.	
2	Screening	**Objective:**	Consensus-based list of preliminary included studies based on content-related criteria
		Method:	Screening of all records using content-related in- and exclusion criteria
		Material:	Abstracts of all articles retrieved
		▪ All hits are screened and evaluated on the basis of the finally consented content-related criteria. ▪ The procedure will be done in two pairs of two reviewers (VLev&YE, VLen&NP). Each pair evaluates half of the hits. ▪ The respective pair of reviewers will meet to discuss and compare their lists of preliminary included studies. ▪ If no consensus is reached, a third reviewer (TM) will read the abstract and moderate the consensus process. ▪ The two lists of preliminary studies (one list per pair) will be joined to one. ▪ If modifications of content-related inclusion / exclusion criteria are necessary, this will be done and be documented.	
3	Eligibility	**Objective:**	Consensus-based list of finally included studies based on content-related criteria, list of excluded studies (incl. reason for exclusion)
		Method:	Full-text screening of the preliminary included studies using content-related in- and exclusion criteria
		Material:	Full text
		▪ All reviewers will read the full text of the preliminary included studies. ▪ Each reviewer creates a list of finally included and a list of excluded studies (with documentation of reason for exclusion). ▪ Consensus will be sought regarding in- and exclusion of studies. The process and decisions will be documented. ▪ The consensus process will result in one list of finally included and one list of excluded studies (with documentation of reason for exclusion).	
4	Inclusion	**Objective:**	Method-specific**, c**onsensus-based lists of finally included studies, lists of excluded studies for each method (incl. reason for exclusion)
		Method:	Full-text screening of the studies included in step 3 using method-specific in- and exclusion criteria
		Material:	Full text
		▪ Method-specific review teams will read the full text of the studies included in step 3 ▪ Each reviewer creates a list of studies to be included on the basis of method-specific criteria (with documentation of reason for exclusion). ▪ Consensus about in- and exclusion will be sought in the method-specific teams. The process and decisions will be documented. ▪ This process will result in two lists of studies to be in- or excluded	

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Critical appraisal of study quality

Noblit & Hare [24] did not take a stance on quality appraisal of the primary studies included in meta-ethnography. Other authors of meta-ethnographic methodological research present contrasting opinions on quality appraisal: Whereas Atkins et al. [27], Campbell et al. [45], Froud et al. [46], Pound et al. [47] or Toye et al. [48] seem to be willing to explore the methodological quality of studies to be included in meta-ethnographies, Britten et al. [30] or France et al. [28] point out, that time could be spent more efficiently by concentrating on the topic of the review [30] or that critical appraisal is redundant as long as it has no implications for in- or exclusion of studies [28]. Thomas & Harden performed critical appraisal of the primary studies included in their thematic synthesis and undertook a sensitivity analysis to determine the impact of low-quality studies on the findings of the review [25]. Overall, the position of qualitative researchers on this topic is still indecisive. What convinced us to include critical appraisal, were the findings of Pindus et al. that a good quality score does not necessarily mean, that the study provides sufficient contextual data [20], and France et al. [28], that poor descriptions of methods do not necessarily equate to a poorly-conducted study. Toye et al. [49] drew attention to the fact, that insightfulness and methodological rigour are different facets of quality and may be quite different. So, we felt that the determination of methodological quality could contribute to the ongoing debate, as long as we go beyond the mere exclusion of methodologically flawed studies [49, 50]. We will follow Thomas & Harden [25] and their sensitivity analysis approach and further will explore the relation between insightfulness and methodological quality in the included studies. To determine the methodological quality of the included studies we chose the widely used Critical Appraisal Skills Programme (CASP) Checklist for qualitative studies [40, 51, 52].

Data extraction and analysis

For meta-ethnography as well as for thematic synthesis researcher pairs will independently extract data into spreadsheets and resolve potential discrepancies by discussion. Unsolved disagreements will be discussed with a fifth researcher. From studies employing mixed methods designs only qualitative data will be extracted. Extracted background data for each study include: study-design, objective, location, setting, participants and method of data collection and analysis.

For the meta-ethnographic analysis second order constructs on subjective needs and desires related to health care will be extracted. Second order constructs are defined as the interpretation of the authors of the primary study [48]. First order constructs (the interpretations of the participants of the primary studies) are to be additionally included in our analysis only if clarification of the second order constructs is required [28]. The second order constructs will lead to third order constructs, in other words: the interpretation of the reviewer team [25, 30]. For the thematic synthesis all of the text in the results/findigs section is to be analyzed in three steps: First, a line-by-line coding, secondly the organization of the resulting codes to descriptive themes and thirdly the development of analytical themes out of the descriptive ones. These analytical themes

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correspond to the third order constructs of the meta-ethnographical analysis [25] and therefore can be used to compare the findings of ME and ThS.

Constant comparison process

Two independent working groups will each conduct a review using one of the outlined methods. Beyond that, our intention is to compare the challenges, questions, and findings arising from each step of the review process to generate empirical material that may clarify the similarities as well as the differences of the two methods. For there is no existing framework that could guide us through the process, we will develop the distinct steps of the comparison during the review process using the method of constant comparison. As an example, we describe how we will proceed if the selection process ends up in two different study samples: To avoid that further differences, particularly the findings of the synthesis in terms of third order constructs / analytical themes, are seen as a result of the divergent study samples, we will analyse both samples with each method. That means: First each group analyses the studies fulfilling the content-related *and* the specific inclusion criteria of 'their' method, and afterwards they include the method-specific studies of the other group – an approach very similar to sensitivity analysis suggested by Thomas & Harden [25]. This will enable us to distinguish between differences resulting from the methodological steps applied and differences based only on the fact, that different studies were included in the QES.

The findings of the two approaches are to be compared with respect to the process of conduction; the resulting third order constructs (ME) / analytical themes (ThS), the scope of the findings and their potential of generalization. Because we expect challenges to occur while performing the particular steps of the review, we decided to determine the exact procedures during the process – a strategy named "Gegenstandsangemessenheit" and seen as an indicator of quality in qualitative research [53–55]. To make this process visible, we will document the respective problem, our discussion and the way we decide to solve it – including the reason for the decision - in a forthcoming methodological article.

A systematic comparison between two qualitative synthesis approaches cannot limit itself to be descriptive, but will have to be interpretive to explain the differences. The point of departure of this analysis will be a comparison of the findings of both approaches. Differences and similarities will be identified. In a second step we will try to understand the differences by looking closer at the respective methods, processes and decisions made during the conduct of the respective study. This will enable us to identify reasons for decisions made and to distinguish between possible reasons that are grounded in the respective methodology or approach vs. other reasons, such as personal backgrounds of the reviewers or special theoretical stances. By doing this we make reference to the comprehensive meta-study approach and plan to explain differences between the findings of the two approaches due to reasons related to data analysis, methods and theories applied [56].

Author background and reflexivity

All participating reviewers are experienced qualitative researchers, with different clinical and scientific backgrounds. NP: psychology and health services research, lecturer for

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qualitative research methods, papers on e.g. qualitative data acquisition and analysis; TM: psychology, rehabilitation sciences and health services research; lecturer in interpretative qualitative methods, papers on e.g. quality criteria of qualitative research and qualitative sampling; VLev/YE: gerontology and palliative care and VLen: Public Health, nursing and physical therapy, lecturer for qualitative and health services research.

This diversity in backgrounds will enable us to adopt a comprehensive view on the topic investigated. Constant discussion and regular meetings will help to avoid skewing the analysis and interpretation and thus minimize reviewer bias. Potential threats to credibility as well as reviewer bias will also be minimized by rigorous and transparent discussion in our interdisciplinary research group "Qualitative Methods in Health Services Research" of the German Network of Health-Services Research, as well as determination and documentation of all procedures during the review process. To disclose how our positions may have influenced the review and whether our attitudes toward the research topic changed during the review, we will include a reflexive statement [35] in the results section of our review.

Discussion

By using two rigorous and codified methods (ME, ThS) for synthesizing qualitative research, we expect to generate in-depth conceptual understanding of what is important for people with poststroke aphasia. The identification of all variations of patients' (living with aphasia) needs and desires related to health care may contribute to tailor health care in a more patient-centred way – which is known to improve therapeutic outcomes [57–59]. The combination of two different QES methods is unique to our study. As far as we know such type of review has never been conducted before and it will allow insights on the impact the QES methods may have not only on the findings but on the whole review process.

A limitation of our study is that we will not include publications in languages other than German or English. This might limit the range of the findings of our QES.

In addition, a comparison of different QES methods can only be made with some methodological limitations. From a pure conceptual or even idealistic perspective, a strict comparison is hardly possible or only possible for two quite similar approaches. We chose two approaches similar enough to answer the same research question (leaving it open to adapt the question during the course of the study if necessary). We assume, however, that the reality of choice over QES approaches is in practice much more dependent on criteria that go beyond methodological considerations – such as personal background and familiarity with a method, personal preferences of team members etc. [21]. In practice, therefore, boundaries between different QES approaches might be blurred with regard to concepts, methods or terms. Since there is no *ideal* comparison in this vein, we find it appropriate to make a few restrictions to the respective QES methodologies in order to allow for substantive comparisons. It is well acknowledged that these restrictions should be an explicit part of the analysis, i.e. the approach to comparison.

Ethics and dissemination

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Ethical approval is not required, because no human subjects or personal data are directly involved in our review. To ensure dissemination, the findings of the QES will be published in an international peer-reviewed journal and presented on national conferences. A distinct method paper will be published in an international peer-reviewed journal.

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Supplementary File 1: Search strategies and relevant references used to test the MEDLINE strategy

Medline (PubMed)

(comprises all terms of "HIRU hedges Qualitative – MEDLINE filter, best balance of sensitivity and specificity" and MeSH and textwords identified during text analysis of twelve relevant references, adding sensitivity)

((Aphasia[mh] OR aphasia*[tw]) AND (Qualitative Research[mh] OR Interviews as Topic[mh] OR Interview[pt] OR Focus Groups[mh] OR experience*[tw] OR qualitative[tw] OR interview*[tw] OR needs[tw] OR themes[tw] OR perspective*[tw]))

PsycINFO (Ovid SP)

1. Aphasia/
2. aphasia*.tw.
3. 1 or 2

[HIRU hedges Qualitative – PsycINFO filter, best balance of sensitivity and specificity]

4. experiences.tw. or interview*.tw. or qualitative*.tw.
5. 3 and 4

Psyndex (PubPsych)

(aphasi* AND (experience* OR qualitativ* OR interview* OR needs OR themes OR perspective*)) AND DB="PSYNDEX"

CINAHL (EBSCO)

1. MH "aphasia+"
2. TI (aphasia*) OR AB (aphasia*)
3. S1 OR S2

[Wilczynski 2007 "small drop sensitivity"-filter]

4. MH "attitude+" OR MH "interviews+" OR MH "qualitative studies"
5. S3 AND S4
6. MH "Interview+" OR MH "Grounded theory" OR MH "Focus Groups" OR MH "Field Studies" OR MH "Life Experiences+"
7. S3 AND S6
8. S5 OR S7

Wilczynski NL, Marks S, Haynes RB. Search strategies for identifying qualitative studies in CINAHL. Qual Health Res. 2007 May;17(5):705-10.

Social Science Citation Index (Web of Science)

((TS=aphasia*) AND (TS=qualitative OR TS=interview*))

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ASSIA (ProQuest)

ALL(aphasia*) AND ALL(qualitative OR interview* OR focus group* OR experience* OR need* OR theme* OR perspective* OR attitude*)

Bielefeld Academic Search Engine (BASE – Advanced Search)

[limited to doctype = Dissertationen und postgraduale Arbeiten]

(aphasia*) AND (qualitative OR interview* OR experience* OR needs OR themes OR perspective*) doctype:183

Twelve relevant references used to test the MEDLINE search strategy

1. Attard MC, Loupis Y, Togher L, Rose ML. Experiences of people with severe aphasia and spouses attending an Interdisciplinary Community Aphasia Group. Disabil Rehabil. 2020 May;42(10):1382-1396.

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7. Burns M, Baylor C, Dudgeon BJ, Starks H, Yorkston K. Asking the Stakeholders: Perspectives of Individuals With Aphasia, Their Family Members, and Physicians Regarding Communication in Medical Interactions. Am J Speech Lang Pathol. 2015 Aug;24(3):341-57.

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ENTREQ-Checklist Item 17-19

		page
Item 1-16		n.a.
17 Coding	Describe the process for coding of data (e.g. line by line coding to search for concepts)	4-5
18 Study comparison	Describe how were comparisons made within and across studies (e.g. subsequent studies were coded into pre-existing concepts, and new concepts were created when deemed necessary).	4-5
19 Derivation of themes	Explain whether the process of deriving the themes or constructs was inductive or deductive	4-5
Item 20-21		n.a.

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PRISMA-P (Preferred Reporting Items for Systematic review and Meta-Analysis Protocols) 2015 checklist: recommended items to address in a systematic review protocol*

Section and topic	Item No	Checklist item	page
ADMINISTRATIVE INFORMATION			
Title:			
Identification	1a	Identify the report as a protocol of a systematic review	1
Update	1b	If the protocol is for an update of a previous systematic review, identify as such	n.a.
Registration	2	If registered, provide the name of the registry (such as PROSPERO) and registration number	2
Authors:			
Contact	3a	Provide name, institutional affiliation, e-mail address of all protocol authors; provide physical mailing address of corresponding author	1
Contributions	3b	Describe contributions of protocol authors and identify the guarantor of the review	11
Amendments	4	If the protocol represents an amendment of a previously completed or published protocol, identify as such and list changes; otherwise, state plan for documenting important protocol amendments	n.a.
Support:			
Sources	5a	Indicate sources of financial or other support for the review	11
Sponsor	5b	Provide name for the review funder and/or sponsor	n.a.
Role of sponsor or funder	5c	Describe roles of funder(s), sponsor(s), and/or institution(s), if any, in developing the protocol	n.a.
INTRODUCTION			
Rationale	6	Describe the rationale for the review in the context of what is already known	2-4
Objectives	7	Provide an explicit statement of the **question(s) the review will address** with reference to participants, interventions, comparators, and outcomes (PICO)	6
METHODS			
Eligibility criteria	8	Specify the study characteristics (such as PICO, study design, setting, time frame) and report characteristics (such as years considered, language, publication status) to be used as criteria for eligibility for the review	6-7
Information sources	9	Describe all intended information sources (such as electronic databases, contact with study authors, trial registers or other grey literature sources) with planned dates of coverage	6-7
Search strategy	10	Present draft of search strategy to be used for at least one electronic database, including planned limits, such that it could be repeated	6, suppl. file 1

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Study records:			
Data management	11a	Describe the mechanism(s) that will be used to manage records and data throughout the review	7-9
Selection process	11b	State the process that will be used for selecting studies (such as two independent reviewers) through each phase of the review (that is, screening, eligibility and inclusion in meta-analysis)	7, Table 2
Data collection process	11c	Describe planned method of extracting data from reports (such as piloting forms, done independently, in duplicate), any processes for obtaining and confirming data from investigators	9
Data items	12	List and define all variables for which data will be sought (such as PICO items, funding sources), any pre-planned data assumptions and simplifications	6, table1
Outcomes and prioritization	13	List and define all outcomes for which data will be sought, including prioritization of main and additional outcomes, with rationale	5
Risk of bias in individual studies	14	Describe anticipated methods for assessing risk of bias of individual studies, including whether this will be done at the outcome or study level, or both; state how this information will be used in data synthesis	n.a.
Data synthesis	15a	Describe criteria under which study data will be quantitatively synthesised	n.a.
	15b	If data are appropriate for quantitative synthesis, describe planned summary measures, methods of handling data and methods of combining data from studies, including any planned exploration of consistency (such as I^2, Kendall's τ)	n.a.
	15c	Describe any proposed additional analyses (such as sensitivity or subgroup analyses, meta-regression)	n.a.
	15d	If quantitative synthesis is not appropriate, describe the type of summary planned	9
Meta-bias(es)	16	Specify any planned assessment of meta-bias(es) (such as publication bias across studies, selective reporting within studies)	n.a.
Confidence in cumulative evidence	17	Describe how the strength of the body of evidence will be assessed (such as GRADE)	n.a.

*** It is strongly recommended that this checklist be read in conjunction with the PRISMA-P Explanation and Elaboration (cite when available) for important clarification on the items. Amendments to a review protocol should be tracked and dated. The copyright for PRISMA-P (including checklist) is held by the PRISMA-P Group and is distributed under a Creative Commons Attribution Licence 4.0.**

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